Commutative Algebra

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0 Preliminaries

Definition 0.1 ([]**.** Ring] In this course a **ring** will mean a *commutative*, *unital* ring (i.e. a commutative ring with a multiplicative identity). There is immediately a non-commutative exception $\mathrm{End}(M) \equiv \{f : M \to M \mid f \text{ is a group homomorphism}\}$ with function composition being the multiplication in this ring.

Definition 0.2. An **R-module** M is an abelian group M with a fixed ring homomorphism

$$\rho : R \to \mathrm{End}(M),$$

we will write $r \cdot m \equiv \rho(r)(m)$. This multiplication has the distributive properties:

1. $r(m_1 + m_2) \equiv \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2)$. Since $\rho(r)$ is an endomorphism of M.
2. $(r_1 + r_2)m \equiv \rho(r_1 + r_2)(m) = \rho(r_1)(m) + \rho(r_2)(m)$. Since ρ itself is a homomorphism of rings.

Examples

- A k-module (where k is a field) gives a vector space
- Every abelian group M is a $\mathbb{Z}$-module in a unique way, via

  $$1_{\mathbb{Z}} \mapsto \mathrm{Id}_M$$

  . This uniquely determines the module structure.
- Every ring R is an R-module.

  $$R \to R$$

  $$r_0 \mapsto (r \mapsto r_0 \cdot r)$$
- $R^{\oplus \mathbb{N}}$ and $R^{\mathbb{N}}$ are R-modules. (Distinguished by the fact that the former must be eventually 0, and the latter need not be).

1 Chain Conditions

Definition 1.1 ([]**.** Noetherian/Artinian] An R-module M is **noetherian** if one of the following hold:

1. Every ascending chain of submodules

   $$M_0 \subseteq M_1 \subseteq \dots$$

   stabilises. That is to say, $\exists k \; \forall j \; M_{k+j} = M_k$. (We call this ACC)

2. Every non-empty set Σ of submodules of M has a maximal element

M is **artinian** if the same holds, but in (1) chains descend instead of ascend (DCC), and in (2) we take minimal instead of maximal

Lemma 1.2.
An R-module is noetherian $\Leftrightarrow$ Every submodule is finitely generated.

This means every Noetherian module is finitely generated, but the converse is false, for example $\mathbb{Z}[T_1, \dots]$ as a module over itself is finitely generated (by 1_R), but not Noetherian as a module or a ring (e.g. $\langle T_1, \dots \rangle$ is not f.g., but is a submodule).

Proof omitted.
Definition 1.3. A ring is **noetherian** (resp. **artinian**) if R as an R-module is noetherian (resp. artinian).
Example 1.4.

- $\mathbb{Z}$ is a noetherian module which is non-artinian. Noetherian since it is a PID, and so the ACC must hold. It is non-artinian since $(2) \supseteq (4) \supseteq (8)$, and this never need terminate.
- $\mathbb{Z}[\frac{1}{2}]/\mathbb{Z} = \{\frac{a}{2^n} + \mathbb{Z} \mid a \in \mathbb{Z}\}$ is an artinian module over $\mathbb{Z}$ which is not noetherian.

However, A ring R is artinian $\Leftrightarrow$ R is noetherian & has *Krull* dimension 0. (More on this later).

Definition 1.5 ([]**.** Exact Sequence] A sequence:

$$\dots \to M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+2} \to \dots$$

of R-module homomorphisms is **exact** if:

$$\forall i \ \operatorname{Im}(f_i) = \ker(f_{i+1})$$

(Slogan: "Two steps get 0").

A **short exact sequence** is an exact sequence of the form

$$0 \to M' \xrightarrow[\text{injective}]{i} M \xrightarrow[\text{surjective}]{} M'' \to 0$$

. This gives

$$M'' \simeq \frac{M}{i(M')}.$$

Where i is some embedding of M' into M.

Lemma 1.6. *Let*

$$0 \to N \to M \to L \to 0$$

be a short exact sequence of R-modules. Then M is noetherian (artinian) iff N And L are noetherian (artinian).

Proof. $L \simeq \dfrac{M}{N}$. This (apparently) does it. Consider module properties and use the lifting-projections trick (I'm pretty sure this does it).

Corollary. *If $M_1, \ldots, M_n$ are noetherian/artinian, then so is:*

$$M_1 \oplus M_2 \oplus \cdots \oplus M_n.$$

Proof. Use the same quotient trick and induct (induce? Do induction.)

Reminder
A module homomorphism

$$M_1 \oplus M_2 \oplus \cdots \oplus M_n \to L$$

is precisely a series of module homomorphisms

$$\phi_i : M_i \to L.$$

Proposition 1.7. *For a noetherian/artinian ring R, every finitely generated R-module is noetherian*

Proof. M finitely generated $\Leftrightarrow \exists n \geqslant 1,\ \exists \phi : R^n \to M$ surjective.

So any quotient of R^n finitely generated ($\Leftarrow$).
If M finitely generated, take generators and build this up to get $\phi : R^n \to M$ ($\Rightarrow$). □

Definition 1.8 ([]**.** R-algebra] An **R-algebra** is a ring A together with a fixed ring homomorphism

$$\rho : R \to A.$$

For example $k \to k[T_1, \ldots, T_n]$ where the first k gets mapped to scalars in the polynomial ring.

We wrtie $r \cdot a = \rho(r)(a)$, so that $\rho(r) = \rho(r) \cdot 1_A = r \cdot 1_A$.

We can view this as a module by 'forgetting' information, e.g. in the example above by forgetting how to multiply polynomials, except by scalars.

In the same way that we talk about modules being finitely generated, we can talk about R-algebras being finitely generated, however the condition is slightly different.

Definition 1.9. An R-algebra is **finitely generated** if

$$\exists n \geqslant 0 \; \exists \phi : R[T_1, \ldots, T_n] \to A,$$

where ϕ is a surjective R-algebra homomorphism. What is an R-algebra homomorphism, you ask?

Definition 1.10. If $\phi : A \to B$ is a map of R-algebras, it is an **R-algebra homomorphism** if ϕ is both a ring homomorphism *and*

$$\phi(r \cdot 1_A) = \phi(r \cdot 1_B).$$

That is to say, the map 'does not touch scalars.'

Lecture 2

Theorem 1.11 ([]**.** *Hilbert's Basis Theorem] Every finitely generated algebra over a noetherian ring is noetherian. For example*

$$k[T_1, \ldots, T_n].$$

Proof. Suffices to prove for $R[T_1, \ldots, T_n]$, since every finitely generated algebra is a quotient of the polynomial ring by (1.9). Moreover, suffices to prove for $R[T]$, since

$$R[T_1, \ldots, T_n] \cong R[T_1, \ldots, T_{n-1}][T_n]$$

so the claim will follow by induction. Let A be an ideal of $R[T]$.

$$\forall i \; A(i) = \{C_0 \,|\, C_0 T^i + \cdots + C_i T^0 \in A, \, C_0, \ldots, C_i \in R\}.$$

Now, each $A(i) \subseteq R$ is an ideal of R, since it is closed under addition, and under multiplication by elements in R, Moreover, we have

$$A(i) \subseteq A(i+1)$$

so this gives an ascending chain. Then since R is noetherian, we know that each $A(i)$ must be a finitely generated ideal, and that the ascending chain of ideals $A(i)$ must eventually stabilise:

$$\exists m \; \forall m' \geqslant m, \; A(m') = A(m)$$

Then we write

$$A(i) = (b_{i,1}, \ldots, b_{i,n_i}), \;\; b_{i,j} \in R$$

Take $f_{i,j} = b_{i,j} \cdot T^i + \text{terms of degree} < i$. Now we define a new ideal B of $R[T]$:

$$B = (f_{i,j})_{\substack{i \leqslant m \\ j \leqslant n_i}}.$$

Then by construction, we can define $B(i)$ similarly to $A(i)$ and obtain that

$$\forall i,\, B(i) = A(i).$$

Also by construction, $B \subseteq A$, since each of the generators of B is contained in A. So now, assume for contradiction that $A \nsubseteq B$. Take $f \in A \setminus B$ of minimal degree i.

It must be the case that $B(i) = A(i)$, so we must be able to find a $g \in B$ such that the leading term of g is the same as f. Then take $f - g$ which has degree $< i$, so this must be in both B and A by the minimality of i. Therefore

$$f = (f - g) + g \in B \ \#$$

□

We can conclude that if R is noetherian, then we have

$$S \subseteq \frac{R[T_1, \ldots T_n]}{I}$$

Then

$$(S) = (S_0),$$

where S_0 is a finite subset of S[1].

2 Tensor Products

2.1 Tensor Products of R-modules

Let M, N be R-modules, then the informal definition of the tensor product is

$$M \bigotimes_R N = \{\sum_{i=1}^{l} m_i \otimes n_i \,|\, m_i \in M,\, n_i \in N\}$$

along with some distributive properties:

- $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$
- $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$
- $(rm) \otimes n = r(m \otimes n) = m \otimes (rn)$

Example 2.1.

[1] We could also have proved this using something called Groebner Bases, but there will be more about those later (I think)

- Consider
$$\frac{\mathbb{Z}}{2\mathbb{Z}} \bigotimes_{\mathbb{Z}} \frac{\mathbb{Z}}{3\mathbb{Z}}$$
has that
$$x \otimes y = (3x) \otimes y = x \otimes (3y) = x \otimes (0 \cdot 0) = 0.$$
So in this case, everything collapses down to nothing, and we have that
$$\frac{\mathbb{Z}}{2\mathbb{Z}} \bigotimes_{\mathbb{Z}} \frac{\mathbb{Z}}{3\mathbb{Z}} = \{0\}$$

- We have that
$$\mathbb{R}^n \bigotimes_{\mathbb{R}} \mathbb{R}^l \simeq \mathbb{R}^{n \cdot l}$$
See later for more.

Recall the definition of bilinearity. When M, N, L all R-modules
$$f : M \times N \to L$$
is R-bilinear if
$$\begin{aligned} &\forall m_0, \;\; n \mapsto f(m_0, n) \\ &\forall n_0, \;\; m \mapsto f(m, n_0) \end{aligned}$$
are both R-linear.

Definition 2.2 ([]**.** Tensor Product of Modules] Let M, N be R-modules.
$$\begin{aligned} \mathcal{F} &= R^{\bigoplus(M \times N)} \\ &= \text{span}_R\{e_{(m,n)} | (m,n) \in M \times N\}. \end{aligned}$$
This is an absolutely *huge* set, with a basis vector for each element of the cartesian product of M and N, so we need to quotient by something. We define $K \subseteq \mathcal{F}$ to be the subset generated by
$$\begin{aligned} &\{(m, n_1) + (m, n_2) - (m, n_1 + n_2) \,|\, m \in M, \;\; n_1, n_2 \in N\} \\ \cup &\{(m_1, n) + (m_2, n) - (m_1 + m_2, n) \,|\, \dots\} \\ \cup &\{r \cdot (m, n) - (rm, n) | \dots\} \\ \cup &\{r \cdot (m, n) - (m, rn) | \dots\} \end{aligned}$$
Then we define
$$M \bigotimes_R N := \frac{\mathcal{F}}{K}.$$

And this gives us an R-bilinear map

$$\begin{aligned} i_{M\otimes N} : M \times N &\to M \otimes_R N \\ (m, n) &\mapsto (m \otimes n) \end{aligned}$$

We call the elements of the form $m \otimes n$ *pure tensors*. The pure tensors generate $M \otimes N$, but not every element is pure (in general).

Proposition 2.3 ([]). *The universal property of a tensor product] The tensor product*

$$\left(M \bigotimes_R N, i_{M\otimes N} \right)$$

satisfies the property that for all R-modules L, and R-bilinear maps $f : M \times N \to L$, there is exactly one R-linear $h : M \otimes N \to L$ such that $f = h \circ i_{M\otimes N}$, i.e. such that the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{i_{M\otimes N}} & M \otimes N \\ & \searrow^{f} & \downarrow h \\ & & L \end{array}$$

Proof. We want to show that

$$h \circ i_{M\otimes N} = f$$

These are two bilinear maps, so let us plug in (m, n).

$$h(i_{M\otimes N}(m, n)) = f(m, n)$$

So we are 'solving' for h, however we already know that there can be at most one option for h, because

$$\{m \otimes n\}_{(m,n)\in M\times N}$$

generate $M \otimes N$ as an R-module.

But why does $h(m \otimes n) = f(m, n)$ extend to an R-linear map

$$h : M \otimes N \to L?$$

Because

$$\begin{aligned} R^{\oplus(M\times N)} &\to L \\ e_{(m,n)} &\mapsto f(m, n) \end{aligned}$$

must vanish on K (because L is an R-module, so we have R-linearity; and moreover this gives us an h from $M \otimes N$ by how we defined the tensor product). Therefore h must exist.

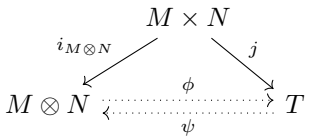
□

Proposition 2.4. *Take M, N to be R-modules and a pair (T, j) where T is an R-module and j is an R-bilinear map, $j : M \times N \to T$ which satisfies the univeral property from 2.3. Then $\exists!$ R-linear isomorphism*

$$\phi : M \otimes_R N \to T$$

such that $\phi \circ i_{M \otimes N} = j$.

Proof. We have



We 'challenge' $M \otimes N$ with the existence of j to get ϕ from the universal property of tensor products.

Then we challenge T with $i_{M\otimes N}$ to get ψ, since T also has the univeral property. We obtain

$$\psi \circ \phi \circ i_{M\otimes N} = i_{M\otimes N}.$$

Therefore $(\psi \circ \phi)$ is the solution we obtain when we challenge $M \otimes N$ with $i_{M\otimes N}$. But so is $\mathrm{Id}_{M\otimes N}$ so it must be the case that $\psi \circ \phi = \mathrm{Id}_{M\otimes N}$

□

Lecture 3

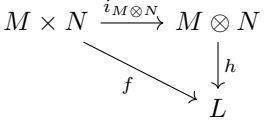
So we have proved that for M, N being R-modules, for all R-modules L, and where $i_{M\otimes N} : M \times N \to M \otimes_R N$:

$$\begin{aligned} \mathrm{Bil}_R(M \times N, L) &\xrightarrow{\sim} \mathrm{Hom}(M \otimes_R N, L) \\ h \circ i_{M\otimes N} &\leftarrow\!\shortmid h \end{aligned}$$

Proposition 2.5. *If M, N R-modules, then*

$$\begin{aligned} \sum m_i \otimes n_i = 0 \Longleftrightarrow &\forall R\text{-module } L, \\ &\forall R\text{-bilinear } f : M \times N \to L, \\ &\sum f(m_i, n_i) = 0. \end{aligned}$$

Proof. Assume $\sum m_i \otimes n_i = 0$. Then f factors through



So therefore $\sum f(m_i, n_i)$ can be written as

$$\begin{aligned} &\sum h(i_{M\otimes N}(m,n)) \\ =&\sum h(m\otimes n) \\ =&h\left(\sum m_i \otimes n_i\right) && \text{(by linearity)} \\ =&0 \end{aligned}$$

If $\sum m_i \otimes n_i \neq 0$, then

$$\sum i_{M\otimes N}(m_i, n_i) = 0$$

since $i_{M\otimes N}(m_i, n_i) = 0$. □

Example 2.6.
Consider $k^m \otimes k^l$, where k is a field. k^m has a basis given by

$$\{e_1, \ldots, e_m\}$$

and k^l has a basis given by

$$\{f_1, \ldots, f_l\}.$$

Then

$$\begin{aligned} k^m \otimes k^l &= \operatorname{span}_k\{v\otimes w \,|\, v \in k^m,\, w \in k^l\} \\ &= \operatorname{span}\{e_i \otimes f_j \,|\, i \leqslant m,\, j \leqslant l\} \end{aligned}$$

So $\{e_i \otimes f_j\}$ forms a basis for $k^m \otimes k^l$.

Proof. Assume $\sum_{i,j} \alpha_{ij} \cdot (e_i \otimes f_j) = 0$. Then $\forall a \leqslant m$, $\forall b \leqslant l$,

$$\begin{aligned} T_{a,b} &: k^m \times k^l \to k \\ T_{a,b}((v_i)_{i=1}^m, (w_j)_{j=1}^l) &= v_a \cdot w_b && (k\text{-bilinear}) \end{aligned}$$

So these maps just map vectors to their multiplied a^{th} and b^{th} coordinates. Then by the proposition above $\sum \alpha_{i,j} \cdot T_{a,b}(e_i, f_j) = \sum \alpha_{i,j} \cdot (e_i \otimes f_j) = 0$ for each a, b. But then

$$\sum \alpha_{i,j} \cdot T_{a,b}(e_i, f_j) = \alpha_{a,b} = 0.$$

So the $e_i \otimes f_j$ do indeed form a basis. □

Example 2.7.
Consider $\mathbb{R}^2 \otimes_{\mathbb{R}} \mathbb{R}^2$, this has infinitely many pure tensors, and has a basis given by:

$$e_1 \otimes f_1,\ e_1 \otimes f_2,\ e_2 \otimes f_1,\ e_2 \otimes f_2.$$

A pure tensor in $\mathbb{R}^2 \otimes \mathbb{R}^2$ looks like

$$(\alpha e_1 + \beta e_2) \otimes (\gamma f_1 + \delta f_2).$$

But then, when we expand brackets, we get

$$(\alpha\gamma)e_1 \otimes f_1 + (\alpha\delta)e_1 \otimes f_2 + (\beta\gamma)e_2 \otimes f_1 + (\beta\delta)e_2 \otimes f_2$$

. These coefficients are *not* generic, since there are linear dependencies between $(\alpha\gamma, \alpha\delta, \beta\gamma, \beta\delta)$. For example

$$1e_1 \otimes f_1 + 2e_1 \otimes f_2 + 3e_2 \otimes f_1 + 4e_2 \otimes f_2$$

is *not* a pure tensor (but if we took other numbers, it might be pure, even though it wouldn't *look* pure, we can manipulate it until it does).

Example 2.8. (Warning)
Consider $\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{2\mathbb{Z}}$: $2 \otimes (1 + 2\mathbb{Z}) = 1 \otimes (2 + 2\mathbb{Z}) = 0$. However, if we consider:

$$2\mathbb{Z} \otimes \frac{\mathbb{Z}}{2\mathbb{Z}}$$

as a submodule, then this submodule contains an element which looks like $2 \otimes (1 + 2\mathbb{Z})$. However, this element is *not* 0. Why?

Indeed, define

$$\begin{aligned} b : (2\mathbb{Z}) \times \frac{\mathbb{Z}}{2\mathbb{Z}} &\to \frac{\mathbb{Z}}{2\mathbb{Z}} \\ b(2n, x + 2\mathbb{Z}) &= nx + 2\mathbb{Z} \\ b(2, 1 + 2\mathbb{Z}) &= 1 + 2\mathbb{Z} \end{aligned}$$

It is notationally tempting to think that the 'natural' embedding of $2\mathbb{Z} \otimes \frac{\mathbb{Z}}{2\mathbb{Z}}$ (and submodules of tensor products in general) is injective, but clearly it is not injective, and there are often things that can be 0 when considered as elements in the first module, which are not 0 in the submodule.

But if $M' \subseteq M$, $N' \subseteq N$ are R-modules and

$$\sum m_i \otimes n_j = 0 \in M' \otimes N',$$

then

$$\sum m_i \otimes n_j = 0 \in M \otimes N.$$

So it *does* go the other way.

Proposition 2.9. *If $\sum m_i \otimes n_i = 0$ in $M \otimes_R N$ then there are finitely generated R-submodules $M' \subseteq M$, $N' \subseteq N$, such that*

$$\sum m_i \otimes n_i = 0 \in M' \otimes_R N'.$$

Idea. First of all, when we write the expression above, we assume that it is meaningful. The idea of the proof is that if it is 0, then there is a finite proof

by playing with the bilinear relationship, so we only see finitely many elements on the left side of the tensor product, and on the right side of the tensor product. So take those elements to generate.

Proof. Let

$$\sum m_i \otimes n_i = 0$$

in

$$M \bigotimes_R N = \frac{R^{\otimes(M\times N)}}{K}.$$

I.e. this sum lies in K, the quotient we use in the construction of $M \otimes N$. Then we know that

$$\sum e_{(m_i,n_i)} \in K.$$

But K is defined with some set of generators, so this is a finite sum:

$$\sum \alpha_i k_i,$$

take the $m'_1, \ldots, m'_n$ which appear in the k_i as generators for M', and similarly for N' to construct these M' and N'.

Corollary. *Let A and B be torsion-free abelian groups. Then*

$$A \bigotimes_{\mathbb{Z}} B$$

is torsion-free as well.

Proof. Assume that

$$n \cdot (\sum a_i \otimes b_j = 0)$$

for some $n \in \mathbb{N}$. Then by the proposition, there must be a finitely generated subgroup[2] given by $A' \subseteq A$ and $B' \subseteq B$ such that

$$n(\sum a_i \otimes b_j) = 0 \in A' \otimes B'.$$

But A' and B' are finitely generated abelian groups, and since A and B are torsion-free, A' and B' must also be torsion-free, but they are finitely generated, so we know the only torsion-free, finitely generated abelian groups are $\mathbb{Z}^n$, so we have

$$A' \otimes B' = \mathbb{Z}^m \otimes \mathbb{Z}^l \simeq \mathbb{Z}^{ml}.$$

This is torsion-free, so $\sum a_i \otimes b_j = 0 \in A' \otimes B' \Rightarrow \sum a_i \otimes b_j = 0 \in A \otimes B$. $\square$

[2]The proposition talks about submodules, but here we are talking about subgroups because they are abelian groups, so deal with it.

Note. $\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^3 \simeq_{\mathbb{C}} \mathbb{C}^6 \simeq_{\mathbb{R}} \mathbb{R}^{12}$. ($\simeq_R$ means isomorphic as R-modules).

But $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}^3 \simeq_{\mathbb{R}} \mathbb{R}^4 \otimes \mathbb{R}^6 \simeq \mathbb{R}^{24}$.

Proposition 2.10 ([]). *Tensor Product Laws]*

$$M \otimes N \xrightarrow{\simeq} N \otimes M.$$

That is to say, they are isomorphic as R-modules. The isomorphism is obvious, we just swap. Moreover

$$(M \otimes N) \otimes P \simeq M \otimes (N \otimes P).$$

One way to prove this is to prove that both of these are isomorphic to $M \otimes N \otimes P$, and also to define what this means (tri-linear maps, and quotients, et cetera et cetera).

Another thing:

$$\left(\bigoplus_i M_i \right) \otimes P \simeq \bigoplus (M_i \otimes P).$$

Yet another thing(!):

$$R \bigotimes_R M \simeq M$$
$$r \otimes m \mapsto rm$$

.

Proof. Example sheet, distributive law is proved in lecture notes.

Corollary. *We have another proof of the tensor-product of free-modules. We will now do it for general R and not just fields (even though our previous proof worked just fine).*

Proof.

$$R^m \otimes_R R^l = \left(\bigoplus_{i=1}^{m} R^i \right) \otimes \left(\bigoplus_{j=1}^{l} R^j \right).$$
$$\simeq \bigoplus_{i,j} R \simeq R^{ml}.$$

□

2.2 Tensor Products of R-linear maps

Proposition 2.11. *For R-linear maps*

$$f : M \to M' \qquad\qquad g : N \to N'$$

There is a unique R-linear map

$$f \otimes g : M \otimes N \to M' \otimes N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Proof. Easy using the universal property. Define an R-bilinear map:

$$\begin{aligned} T : M \times N &\to M' \otimes N' \\ T(m, n) &\mapsto f(m) \otimes g(n) \end{aligned}$$

Then we are immediately done by the universal property of $M \otimes N$, since this gives us $f \otimes g$ as in the statement. □

Lecture 4

Let

$$\begin{aligned} f &: M \to M' \\ g &: N \to N' \end{aligned}$$

be R-linear maps, then we define $f \otimes g$ as:

$$\begin{aligned} f \otimes g &: M \otimes N \to M' \otimes N' \\ (f \otimes g)(m \otimes n) &= f(m) \otimes g(n) \end{aligned}$$

Exercise: Show that $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (g \circ i)$

Example: Consider

$$\begin{aligned} T &: k^a \to k^b \\ S &: k^c \to k^d. \end{aligned}$$

Then

$$\begin{aligned} T \otimes S : k^a \otimes k^c &\to k^b \otimes k^d \\ k^{ac} &\to k^{bd}, \end{aligned}$$

This is characterised on the basis of k^{ac} via

$$\begin{aligned} (T \otimes S)(e_i \otimes e_j) &= T(e_i) \otimes S(e_j) \\ &= \sum_{l,t} [T]_{l,i} \, [S]_{t,j} \, (f_l \otimes f_t). \end{aligned}$$

We order the basis of $k^a \otimes k^c$:

$$\begin{matrix} e_1 \otimes e_1, & \dots & e_1 \otimes e_c, \\ e_2 \otimes e_1, & \dots & e_2 \otimes e_c, \\ \vdots & & \vdots \\ e_a \otimes e_1, & \dots & e_a \otimes e_c. \end{matrix}$$

And we order the range similarly. Then we can write

$$[T \otimes S] = \begin{pmatrix} [T]_{1,1}S & [T]_{1,2}S & \dots & [T]_{1,a}S \\ \vdots & \vdots & & \vdots \\ [T]_{b,1}S & [T]_{b,2}S & \dots & [T]_{b,a}S \end{pmatrix}$$

This is called the Kronecker product of matrices.

Proposition 2.12. *Let*

$$f : M \to M'$$
$$g : N \to N'$$

be R-linear functions, then

1. *If f, g are isomorphims, so is $f \otimes g$.*
2. *If f, g are surjective, so is $f \otimes g$.*

Proof.

1. $f^{-1} \otimes g^{-1}$ is a two-sided invese for $f \otimes g$. (Just apply the result from the exercise)
2. We have that $\mathrm{Im}(f \otimes g)$ contains every pure tensor of $M' \otimes N'$. Then $f \otimes g$ is surjective, and we are done.

□

There can be no injectivity claim here, as we saw before (but will repeat here), we have:

$$f : \mathbb{Z} \xrightarrow{\cdot p} \mathbb{Z}$$
$$g : \frac{\mathbb{Z}}{p\mathbb{Z}} \xrightarrow{\mathrm{Id}} \frac{\mathbb{Z}}{p\mathbb{Z}}$$

Then

$$\begin{aligned} (f \otimes \mathrm{Id}_{\frac{\mathbb{Z}}{p\mathbb{Z}}})(a \otimes b) &= pa \otimes b \\ &= p(a \otimes b) \\ &= a \otimes pb \\ &= a \otimes 0 \end{aligned}$$

So this is the zero map, but the ring is not the zero ring, so this map is not injective, despite each of its tensor components being injective.

2.3 Tensor Products of R-algebras

Let B, C be R-algebras (rings with a fixed ring homomorphism).

$$\phi_B : R \to B$$
$$\phi_C : R \to C$$

We can forget structure, and make B and C into R-modules, if we want to. Then we can consider

$$B \otimes_R C$$

as a tensor product of modules, and try to make it into a ring, and then into an algebra.

First of all, we want:

$$(b \otimes c) \cdot (b' \otimes c') = (b \cdot b') \otimes (c \cdot c')$$

We need to check that this is indeed well-defined:

Fix $(b, c) \in B \times C$, then we have an R-bilinear map:

$$B \times C \to B \otimes C$$
$$(b', c') \mapsto (bb') \otimes (cc')$$

Then from the universal property of tensor products, this R-bilinear map gives rise to an R-linear map:

$$B \otimes C \to B \otimes C$$
$$b' \otimes c' \to bb' \otimes cc'.$$

So this multiplication is well-defined. □

Showing that this satisfies the ring axioms is an easy check. Then the R-algebra structure given by

$$\begin{aligned} R &\to B \otimes C \\ r &\mapsto (r \cdot 1_B) \otimes 1_C \\ &\mapsto 1_B \otimes (r \cdot 1_C) \\ &\mapsto r \cdot (1_B \otimes 1_C) \end{aligned}$$

Gives us the fixed ring homomorphism.

Example: There exists an isomorphism, φ, with:

$$\varphi : R[X_1, \ldots, X_n] \otimes R[T_1, \ldots, T_r] \xrightarrow{\sim} R[X_1, \ldots, X_n, T_1, \ldots, T_r].$$

(N.B. We denote the tensor product of algebras the same way as the tensor product of modules, just remember that there is extra structure when we tensor algebras).

Proof. First of all, we have a basis for the left-hand side, given by

$$\{a \otimes b\}$$

such that $a \in R[X_1, \dots X_n]$, $b \in R[T_1, \dots, T_r]$, where a and b are both monomials.

Moreover, we also have a basis for the right-hand side, given by

$$\{ab \,|\, a \text{ a monomial, } b \text{ a monomial}\}.$$

Now, define $\varphi(a, b) = ab$, which gives an R-module isomorphism. So we need to check that this is an R-algebra homomorphism.

$$\varphi(r \otimes 1) = r \cdot 1 \in R[X_1, \dots, X_n, T_1, \dots, T_r]$$

gives a ring homomorphism. Then we need to check that we have a module homomorphism:

$$\begin{aligned}
\varphi\left((\sum_i p_i \otimes q_i)(\sum_j h_j \otimes g_j)\right) &= \varphi\left(\sum_{i,j} p_i h_j \otimes q_i g_j\right) \\
&= \sum_{i,j} (p_i h_j) \cdot (q_i g_j) \\
&= \sum_{i,j} (p_i q_i)(h_j g_j) \\
&= \left[\sum_i \varphi(p_i \otimes q_i)\right] \cdot \left[\sum_j \varphi(h_j \otimes g_j)\right] \\
&= \varphi\left(\sum_i p_i \otimes q_i\right) \cdot \varphi\left(\sum_j h_j \otimes g_j\right).
\end{aligned}$$

In general

$$\frac{R[X_1, \dots, X_n]}{I} \bigotimes \frac{R[T_1, \dots, T_r]}{J} = \frac{R[X_1, \dots, X_n] \otimes R[T_1, \dots, T_r]}{L}$$

where L is some ideal that we can construct somehow (unspecified), however, when working with R-algebras, we have that these things are

$$\simeq \frac{R[X_1, \dots, X_n, T_1, \dots, T_r]}{(I^e + J^e)}.$$

where I^e is the extension of I, that is to say, the ideal generated by elements of I but considered as an ideal of the larger ring.

Example: Consider

$$\frac{\mathbb{C}[X,Y,Z]}{(f,g)} \bigotimes_{\mathbb{C}} \frac{\mathbb{C}[W,U]}{(h)}$$

This is isomorphic, as a $\mathbb{C}$-algebra, to

$$\frac{\mathbb{C}[X,Y,Z,W,U]}{(f,g,h)}.$$

Proposition 2.13 ([]). *Universal property of tensor product of algebras] Let A, B be R-algebras, then*

1. *For all R-algebras C, for all algebra maps*

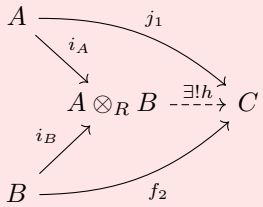
$$f_1 : A \to C$$

$$f_2 : B \to C,$$

there exists a unique

$$h : A \otimes B \to C$$

such that



commutes, i.e. $f_1 = h \circ i_A$, and $f_2 = h \circ i_B$.

2. *This characterises $(A \otimes B, i_A, i_B)$ uniquely. (But we won't prove this).*

Proof. $A \otimes B$ is an R-algebra, which is generated by the set:

$$\{a \otimes 1 \,|\, a \in A\} \cup \{1 \otimes b \,|\, b \in B\}.$$

This would not generate $A \otimes B$ as an R-module, but since we have the additional structure of an algebra, we can multiply elements together to obtain everything else. This immediately implies the *uniqueness* of h, because h must send these elements 'where they get sent' (so long as h exists).

To show existence consider the R-bilinear map:

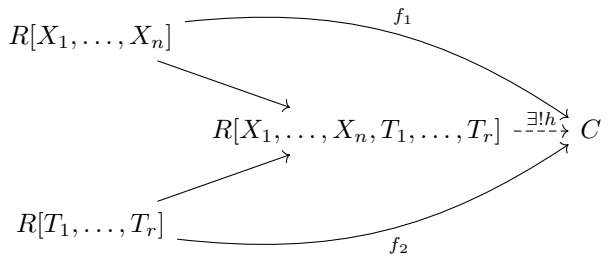
$$\begin{aligned} A \times B &\to C \\ (A,B) &\mapsto f_1(a) f_2(b) \end{aligned}$$

Then we use the universal property of tensor products of R-modules to guarantee that there exists an R-linear

$$h : A \otimes B \to C$$

. Then we need only check h is a homomorphism, just as we did with φ earlier. □

Example: In the case of the algebra given by the polynomial ring:



Where

$$R[X_1, \ldots X_n, T_1, \ldots, T_r] \simeq R[X_1, \ldots, X_n] \otimes R[T_1, \ldots, T_r]$$
$$\text{product of monomials} \leftarrow\!\shortmid \text{monomial } \otimes \text{ monomial.}$$

Lecture 5

Two comments:

(1) If

$$f : A \to A'$$

and

$$g : B \to B'$$

are R-algebra homormorphisms, then

$$f \otimes g : A \otimes B \to A' \otimes B'$$

is an R-algebra homomorphism.

(2) R-algebra homomorphisms:

$$\frac{R}{I} \otimes \frac{R}{J} \simeq \frac{R}{I+J}$$
$$A \otimes B \simeq B \otimes A$$
$$A \otimes (B \times C) \simeq (A \otimes B) \times (A \otimes C)$$
$$\Rightarrow A \otimes B^n \simeq (A \otimes B)^n$$
$$(A \otimes B) \otimes C \simeq A \otimes (B \otimes C).$$

Just like in the example sheet, but check that they satisfy the ring homomorphism property.

3 Restriction and extension of scalars

3.1 Modules

We have two rings R and S in the background, and some specific ring homormophism:

$$f : R \to S.$$

Then if M is an S-module, then it is also an R-module. (The example to have in mind is that a vector space over $\mathbb{C}$ is a vector space over $\mathbb{R}$ with double the dimension).

Consider

$$r \cdot m = f(r) \cdot m \; r \in R, \, m \in M.$$

How do we verify this gives us the structure of a module, well we have a map

$$R \xrightarrow{f} S \to \operatorname{End} M.$$

Then because of our definition of a module, we know that M must be an S module, and don't need to start checking distributivity and other annoying axioms.

Example 3.1. If

$$f : \mathbb{R} \hookrightarrow \mathbb{C}$$

then $\mathbb{C}^n$ is a $\mathbb{C}$-module means that it is also an $\mathbb{R}$-module:

$$\mathbb{C}^n \simeq \mathbb{R}^{2n}.$$

3.2 Extension of scalars

We still have $f : R \to S$ a ring homomorphism, and M an S-module which is also an R-module, and we have N which is an R-module. Then from this, we can form

$$M \bigotimes_R N$$

which is an R-module, since we forget the action of S on M in order to form this tensor product. The restriction of scalars tells us that $M \otimes_R N$ is *also* an S-module.

We define

$$s \cdot (m \otimes n) := (sm) \otimes n.$$

Is this well-defined? Yes. We take an R-bilinear map:

$$M \times N \to M \otimes N$$
$$\text{via } (m, n) \mapsto (sm) \otimes n$$

Then by the universal proprty, we obtain an

$$h_s : M \otimes_R N \to M \otimes_R N$$

such that $h_s(m \otimes n) = (sm) \otimes n$. To check that this is in fact a module, I define a function

$$\phi : S \to \mathrm{End}(M \otimes_R N)$$
$$\phi(s) = h_s$$

Check that $h_s \in \mathrm{End}(M \otimes_R N)$ and that ϕ is a ring homomorphism.

Then, if we start treating $M \otimes_R N$ as an S-module "out of nowhere," then this is the default construction we will be using.

Examples 3.2. (1) We have that

$$S \otimes_R R \simeq S$$

as R-modules. This sends

$$s \otimes r \mapsto s \cdot f(r)$$

where f is the ambient homomorphism we're working with. This is also S-linear, and it's enough to check this on pure tensors:

$$s'(s \otimes r) \mapsto (s's\cdot)$$
$$= (s's) \otimes r = s'(s \cdot f(r))$$

So $s'(s \cdot f(r))$ is the image of $s \otimes r$ under this map. Therefore, the clear example is that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R} \simeq \mathbb{C}$ as $\mathbb{C}$-modules.

(2) If M is an S-module and N_i are R-modules, then

$$M \otimes (\bigoplus_i N_i) = \bigoplus_i (M \otimes N_i)$$

as S-modules. (We turn each of the $M \otimes N_i$ into S-modules, and then direct sum over S-modules).

The concrete example would be that

$$\mathbb{C} \otimes \mathbb{R}^n \cong \mathbb{C}^n$$

as $\mathbb{C}$-modules.

(3) Take $\mathbb{C}^n$ as a $\mathbb{C}$-module, then restrict to $\mathbb{R}$, and get:

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

as $\mathbb{R}$-modules. Then extend to $\mathbb{C}$ and get

$$\mathbb{C} \otimes \mathbb{R}^{2n} \cong \mathbb{C}^{2n}.$$

So we do lose information by going to $\mathbb{R}$, since $\mathbb{C} \otimes \mathbb{C}^n$ does not give us this, so these operations are not inverse.

(4) Let $\mathbb{R}^n$ be an $\mathbb{R}$-module, then we can extend and get

$$\mathbb{C} \otimes_R \mathbb{R}^n \cong \mathbb{C}^n$$

over $\mathbb{C}$. Or we can restrict and get

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

over $\mathbb{R}$.

(5) Let $\mathbb{Z}^n$ be a $\mathbb{Z}$-module, and take

$$f : \mathbb{Z} \to \frac{\mathbb{Z}}{2\mathbb{Z}}.$$

Then extend this to $\frac{\mathbb{Z}}{2\mathbb{Z}}$, and get

$$\frac{\mathbb{Z}}{2\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Z}^n \cong \left(\frac{\mathbb{Z}}{2\mathbb{Z}}\right)^n.$$

This is also called an *extension of scalars*, even though it might not look like an extension.

Example 3.3. Take $\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^l$, then this is, in one way:

$$\mathbb{C}^n \otimes_R \mathbb{R}^l \cong \mathbb{R}^{2n} \otimes_{\mathbb{R}} \mathbb{R}^l \cong \mathbb{R}^{2nl} \cong \mathbb{C}^{nl}$$

all as modules over $\mathbb{R}$, where the last isomorphism comes from equality of dimensions.

Another way: We can also view

$$\begin{aligned} \mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^l &\cong \mathbb{C} \otimes_{\mathbb{C}} (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^l) \\ &\cong \mathbb{C}^n \otimes_{\mathbb{C}} \mathbb{C}^l \\ &\cong \mathbb{C}^{nl} \end{aligned}$$

So, when we prove the proposition we are about to prove, the construction will go like:

$$v \otimes u \mapsto v \otimes (1 \otimes u) \mapsto v \otimes u$$

This is strange, because we are getting 'more room' on the pure tensors, but the key point is that when we tensor over $\mathbb{C}$ we get more non-pure tensors than we had originally. So it is wrong to only look at pure tensors.

Proposition 3.4. *If M is an S-module, and N is an R-module, then we have*

$$M \otimes_R N \cong M \otimes_S (S \otimes_R N)$$

as S-modules, via

$$\begin{aligned} m \otimes n &\mapsto m \otimes (1 \otimes n) \\ (sm) \otimes n &\leftarrow\!\shortmid m \otimes (s \otimes n) \end{aligned}$$

Proof. In example sheet, or lecture notes.

Proposition 3.5. *Let M, M' be S-modules, and N, N' be R-modules, then we have:*

1. *An S-module isomorphism:*

$$M \otimes_R N \cong N \otimes_R M$$

   via

$$m \otimes n \leftrightarrow n \otimes m$$

2. *An S-module isomorphism:*

$$(M \otimes_R N) \otimes_R N' \cong M \otimes_R (N \otimes N').$$

   So we get associativity, on the LHS we extend to S-modules twice (inside and outside the brackets) and on the RHS we need to extend to an S-module only once.

3. *Similarly to above, we have*

$$(M \otimes_R N) \otimes_S M' \cong M \otimes_S (N \otimes_R M')$$

4. *We also have the distributive property:*

$$M \otimes_R (\bigoplus_i N_i) \cong \bigoplus_i (M \otimes N_i)$$

Proof. We will prove only the third claim, since all the proofs are very similar. We have

$$(M \otimes_R N) \otimes_R M' \cong (M \otimes_S (N \otimes_R S)) \otimes_S M'.$$

Now this is just a tensor product of 3 S-modules, so we can just use ordinary tensor product associativity to get

$$M \otimes_S ((N \otimes_R S) \otimes_S M').$$

But then, we don't need to "prepare" the module N in this expression, just like the first expression, so we get

$$M \otimes_S (N \otimes_R M').$$

□

For a concrete example, we have

$$\begin{aligned} &\mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R}^l \otimes \mathbb{R}\mathbb{R}^n) \\ \cong &(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^l) \otimes_{\mathbb{C}} (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n). \\ \cong &\mathbb{C}^l \otimes_{\mathbb{C}} \mathbb{C}^n \cong \mathbb{C}^{nl} \end{aligned}$$

Corollary 3.6. *Let N, N' be R-modules, then*

$$S \otimes_R (N \otimes_R N') \cong (S \otimes_R N) \otimes_S (S \otimes_R N')$$

Proof. "One line[3]":

$$\begin{aligned} S \otimes_R (N \otimes_R N') &\cong (S \otimes_R N) \otimes_R N' && \text{(by 2. in prop 3.5)} \\ &\cong (S \otimes_R N) \otimes_S (S \otimes_R N') && \text{(doing the "preparation")} \end{aligned}$$

□

Lecture 6 So for example, we have

$$\begin{aligned} \mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R}^m \otimes_{\mathbb{R}} \mathbb{R}^l) &\cong (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^m) \otimes_{\mathbb{C}} (\mathbb{C} \otimes \mathbb{R}^l) \\ &\cong \mathbb{C}^m \otimes \mathbb{C}^l \cong \mathbb{C}^{ml} \end{aligned}$$

By induction with the corollary, we can conlude that

$$S \otimes_R (N_1 \otimes_R \cdots \otimes_R N_l) \cong (S \otimes_R N_1) \otimes_S \cdots \otimes_S (S \otimes_R N_l).$$

And, in the example sheet it is proved that we can basically move the brackets around, since the tensor product is associative.

[3]Lecturer "lines" to real life lines apparently have a 1 to 2 conversion rate, although to be fair, he *just* managed to fit it on one.

3.3 Extension of scalars on morphisms

Let $f : N \to N'$, be R-linear, and let N, N' be R-modules, and let M be an S-module. then:

$$\mathrm{Id}_M \otimes f : M \otimes_R N \to M \otimes_R N',$$

where this map is, by construction, an R-linear map. But in fact it is also S-linear, which we will verify:

$$\begin{aligned}(\mathrm{Id}_M \otimes f)(s(m \otimes n)) &= (id_M \otimes f)((sm) \otimes n) \\ &= (sm) \otimes f(n) \\ &= s((\mathrm{Id}_M \otimes f)(m \otimes n))\end{aligned}$$

If we have

$$T : \mathbb{R}^n \to \mathbb{R}^l$$

an $\mathbb{R}$-linear map, where $\mathbb{R}^n$ has basis $\{e_1, \ldots, e_n\}$, and $\mathbb{R}^l$ has basis $\{f_1, \ldots, f_l\}$. Then if we consider

$$\mathrm{Id}_{\mathbb{C}} \otimes T : \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n \to \mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^l$$

where these modules have bases given by $1 \otimes e_i$ and $1 \otimes f_j$ respectively. Then we want to see what this map does on basis elements, and so we write:

$$\begin{aligned}(\mathrm{Id}_{\mathbb{C}} \otimes T)(1 \otimes e_i) &= (1 \otimes T(e_i)) \\ &= 1 \otimes (\sum_{j=1}^{l} [T]_{ji} \cdot f_j) \\ &= \sum_{j=1}^{l} [T]_{ji} \cdot (1 \otimes f_j)\end{aligned}$$

So $[\mathrm{Id}_{\mathbb{C}} \otimes T]$ is the same as $[T]$ as a matrix[4]

3.4 Extension of scalars on algebras

Let A, B be R-algebras, then we have the construction of the tensor product of algebras $A \otimes_R B$ is an R-algebra. Now, it is quite easy to see that $A \otimes_R B$ is both an A-algebra and a B algebra, by sending

$$\begin{aligned}A &\to A \otimes_R B \\ a &\mapsto a \otimes 1\end{aligned}$$

$$\begin{aligned}B &\to A \otimes_R B \\ b &\mapsto 1 \otimes b\end{aligned}$$

[4]So don't be afraid of the extension of scalars on morphisms.

Example 3.7. If we have our ambient function $f : R \to S$, and $R[X_1, \ldots, X_n]$ is a polynomial ring over R, then

$$S \otimes R[X_1, \ldots, X_n] \cong S[X_1, \ldots, X_n],$$

both S-algebras.

Proof. We already have an isomorphism between these as S-modules, which is given by

$$\varphi(s \otimes p) = sp$$
$$\varphi(1 \otimes 1) = 1$$

Now we just need to verify that multiplication is preserved (we won't do it, but you argue that multiplication is preserved for pure tensors, and then argue that this is enough for the whole algebra).

We can do something a little more complicated, and have

$$S \otimes \left(\frac{R[X_1, \ldots, X_n]}{I} \right) \cong \frac{S[X_1, \ldots, X_n]}{I^e}$$

where I^e is the ideal of $S[X_1, \ldots, X_n]$ generated by $f(I)$.

Two more propositions which we aren't going to prove:

Proposition 3.8. *If A is an R-algebra, and B is an S-algebra, then*

$$A \otimes_R B \cong (A \otimes_R S) \otimes_S B$$

as S-algebras. This is what we called previously the "preparation".

Proof. We already know this for modules, so just check that the identity is sent to the identity, and multiplication is preserved. Use the ring homomorphism f from R to S which we have been assuming exists.

Proposition 3.9. *Let A and B be R-algebras, then*

$$S \otimes_R (A \otimes_R B) \cong (S \otimes_R A) \otimes_S (S \otimes_R B)$$

as S-algebras.

Proof. Just as above, we know it for modules, and then check it works for algebras too.

4 Exactness

4.1 Exactness properties of the tensor products of a fixed module

Let M be an R-module, we will define $T_M(N) = M \otimes_R N$ where N is any R-module. If $f : N \to N'$ is R-linear homomorphism of R-modules, then we also have

$$T_M(f) = \mathrm{Id}_M \otimes f$$

so T_M is defined not just on modules, but also on homomorphisms.

Our first goal is to show that if

$$A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

is an exact sequence of R-modules (i.e. g is surjective, and f has image the kernel of g, but does not need to be injective since we don't have the starting 0). Then

$$M \otimes_R A \xrightarrow{T_M(f)} M \otimes_R B \xrightarrow{T_M(g)} C \to 0.$$

is also exact. T_M acts on both the objects and the morphisms of the category, and preserves composition and identity, so it is a functor from $\mathbf{Module}_R$ to $\mathbf{Module}_R$.

Definition (Right Exact)**.** The property of preserving these sorts of exact sequences (with a 0 at the end but not at the start) is called being **right exact**. T_M therefore is right exact.

Left exactness is exactly what you would think, and they are not identical concepts, and in fact T_M is *not* left exact.

Definition 4.1 (The Hom functor)**.** If Q and P are R-modules, then

$$\mathrm{Hom}_R(Q, P) = \{f : Q \to P \mid f \text{ is } R\text{-linear } \}.$$

In fact, $\mathrm{Hom}_R(Q, P)$ is itself an R-module via

$$(r \cdot \varphi)(q) = r \cdot (\varphi(q)).$$

Then we can generate two **Hom functors** from this construction:

$$\mathrm{Hom}_R(Q, \cdot)$$

and

$$\mathrm{Hom}_R(\cdot, P),$$

where Q and P are both *fixed* R-modules in these definitions.

If $f : N' \to N$ is an R-linear map, then we can apply the Hom functor to both of them, and also to f, to get

$$\begin{aligned} \mathrm{Hom}_R(Q, N') &\xrightarrow{\mathrm{Hom}_R(Q,f)} \mathrm{Hom}_R(Q, N) \\ \varphi &\longmapsto f \circ \varphi := f_*(\varphi) \end{aligned}$$

Obviously, we have the other Hom functor, which works like:

$$\begin{aligned} \mathrm{Hom}_R(N, P) &\xrightarrow{\mathrm{Hom}_R(f,P)} \mathrm{Hom}_R(N', P) \\ \varphi &\mapsto \varphi \circ f := f^*(\varphi) \end{aligned}$$

Proposition 4.2.

1. *If*
$$0 \to A \xrightarrow{f} B \xrightarrow{g} C$$
is exact, then so is
$$0 \to \mathrm{Hom}_R(Q, A) \to \mathrm{Hom}_R(Q, B) \to \mathrm{Hom}_R(Q, C)$$
with the right morphisms defined.
2. *If*
$$A \xrightarrow{f} B \xrightarrow{g} C \to 0$$
is exact, then so is
$$0 \to \mathrm{Hom}_R(C, P) \to \mathrm{Hom}_R(B, P) \to \mathrm{Hom}_R(A, P).$$

Both of these are Left exact because at the end, they start with a 0*, but for claim 2, we need to start with a* 0 *on the right, since the second functor is order-reversing.*

Proof. In example sheets, but there is only one thing you can really do, you do it, and then it works.

Lemma 4.3 (Using Hom functors to prove exactness)**.** *Consider some sequence of* R*-modules (not necessarily exact),*

$$A \xrightarrow{f} B \xrightarrow{g} C, \tag{*}$$

and assume that for all R-modules P, we have

$$\mathrm{Hom}_R(C,P) \to \mathrm{Hom}_R(B,P) \to \mathrm{Hom}_R(A,P)$$

is exact. Then so is (*).

Proof. Step 1: Take $P = C$, then we get

$$\mathrm{Hom}_R(C,C) \to \mathrm{Hom}_R(B,C) \to \mathrm{Hom}_R(A,C)$$

which we know is exact, so we can take

$$id_C \mapsto \mathrm{Id}_C \circ g = g \mapsto \mathrm{Id}_C \circ g \circ f = g \circ f.$$

But then $g \circ f = 0$, since going two steps in an exact sequence always gives 0. Therefore $\mathrm{Im}(f) \subseteq \ker(g)$.

Step 2: Take $P = \frac{B}{\mathrm{Im}\, f}$[5]. Then we have

$$\mathrm{Hom}_R(C, \frac{B}{\mathrm{Im}\, f}) \to \mathrm{Hom}_R(B, \frac{B}{\mathrm{Im}\, f}) \to \mathrm{Hom}_R(A, \frac{B}{\mathrm{Im}\, f}).$$

Then we use the other property of exactness, which is that if we start somewhere that goes to 0, then it has a preimage. So take

$$h : B \to \frac{B}{\mathrm{Im}\, f}$$

to be the canonical quotient map of B to the quotient of B by $\mathrm{Im}\, f$. Then we get, starting at the middle,

$$h \mapsto h \circ f = 0,$$

since f takes you first to $\mathrm{Im}\, f$, and then h sends you to 0. So we have a preimage of h which we call e, and we have:

$$e \mapsto e \circ g = h.$$

So $e \circ g = h$, so $\ker g \subseteq \ker h$. So $\ker(g) \subseteq \mathrm{Im}(f)$. □

Now, consider that

$$\mathrm{Hom}_R(M \otimes_R N, L) \leftrightarrow \mathrm{Bil}(M \times N, L)$$

by the universal property. Then we can also say

$$\mathrm{Hom}_R(M \otimes_R N, L) \leftrightarrow \mathrm{Bil}(M \times N, L) \leftrightarrow \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$

[5] What we call the *Cokernel* of f.

Where the inner Hom functor is R-linear, and then the outer Hom functor must be R-linear, since for every fixed m we need to make sure we get a R-linear map in N, and then this assignment from m to these maps is *itself* *Lecture 7* R-linear. This map is given by

$$\varphi \mapsto (n \mapsto (m \mapsto \varphi(m \otimes n))).$$

Then we claim the following

Proposition 4.4. *Let M be an R-module, then*

$$M \otimes_R (\,\cdot\,) = T_M(\,\cdot\,)$$

is right exact.

Proof. Take an exact sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \to 0,$$

and take an arbitrary R-module P. Then apply the functor $\mathrm{Hom}_R(\,\cdot\,, P)$, and *then* apply the functor $\mathrm{Hom}_R(M, \,\cdot\,)$. Doing this, we obtain first

$$0 \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(C, P)) \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(B, P)) \to \mathrm{Hom}_R(M, \mathrm{Hom}_R(A, P)).$$

Which is exact, since Hom functors are left exact. But then we have the bijection between Hom applied to the tensor product and the Hom functor applied twice (see above), and therefore have

$$0 \to \mathrm{Hom}(M \otimes_R C, P) \to \mathrm{Hom}_R(M \otimes_R B, P) \to \mathrm{Hom}_R(M \otimes_R A, P).$$

And moreover, the bijections between these two short exact sequences form a commutative diagram, so this second exact sequence is exact. Now recall lemma 4.3, and apply this twice to the sequence above (since P was arbitray). Therefore we know that

$$0 \to M \otimes_R C \to M \otimes_R B \to M \otimes_R A.$$

□

Warning: We proved that

$$A \to B \to C (\to 0)$$

remains exact when we tensor on the right with M, but that does *NOT* imply that

$$A \to B \to C$$

remains exact when we tensor. For example, we could take

$$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$$

and tensor with $\mathbb{Z}/2\mathbb{Z}$, to get

$$0 \to \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z},$$

which is isomorphic to

$$0 \to \mathbb{Z}/2\mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}/2\mathbb{Z},$$

which is not exact.

4.2 Flat Modules - A first encounter

Definition 4.5 (Flat Module)**.** An R-module M is **Flat** if, whenever we have

$$f : N \to N'$$

an R-linear injective map, then

$$\mathrm{Id}_M \otimes f : M \otimes_R N \to M \otimes_R N'$$

is injective.

Examples 4.6.

1. $\mathbb{Z}/2\mathbb{Z}$ is not a flat $\mathbb{Z}$-module, as we just saw.
2. Free modules are *always* flat. To see this, consider

   $$f : N \to N',$$

   an injective R-linear map. Then we tensor with $R^{\oplus I}$ where I is some indexing set, to get

   $$\begin{array}{ccc} R^{\oplus I} \otimes_R N & \xrightarrow{\mathrm{Id}_{R^{\oplus I}} \otimes f} & R^{\oplus I} \otimes_R N' \\ \downarrow \simeq & & \downarrow \simeq \\ N^{\oplus I} & \xrightarrow{(n_i)_{i\in I} \mapsto (f(n_i))_{i \in I}} & N'^{\oplus I} \end{array}$$

   The bottom arrow is injective, since we apply an injective map coordinate by coordinate. We also have that the diagram commutes, so the top arrow is also injective, and therefore $R^{\oplus I}$ is flat.
3. The base ring matters! $\mathbb{Z}/2\mathbb{Z}$ is a free $\mathbb{Z}/2\mathbb{Z}$-module. Therefore it is a flat $\mathbb{Z}/2\mathbb{Z}$-module.

4. An R-module M is torsion-free if: $\forall r \in R : r \nmid 0$; $\forall m \in M : m \neq 0$ we have $rm \neq 0$. Then we have

   Claim: Flat modules are torsion-free.

   Proof. Take M not torsion free, then

   $$\exists r_0 \in R : r \nmid 0$$

   $$\exists m \in M : m \neq 0.$$

   And therefore $r_0 m_0 \neq 0$. But now we have a map

   $$\begin{aligned} f : R &\to R \\ r &\mapsto r_0 r. \end{aligned}$$

   Then $\ker(f) = \{0\}$, since r_0 not a zero-divisor. Therefore f is injective. Then we have

   $$\begin{array}{ccc} M \otimes_R R & \xrightarrow{\mathrm{Id}_M \otimes f} & M \otimes_R R \\ \downarrow \simeq & & \downarrow \simeq \\ M & \xrightarrow{m \mapsto r_0 m} & M \end{array}$$

   commutes. Then the bottom map has $0 \neq m_0$ in the kernel, so the top arrow cannot be injective.

   Special Case: Let R be an integral domain (A non-zero ring with no zero-divisors, or equivalently the 0 ideal is prime) and let $0 \neq I$ be an ideal of R. Then

   $$R/I \text{ not flat.}$$

   Proof. If $I = R$ then $R/I = 0$ which is not flat.

   Now assume that $I \subsetneq R$. Take $x \in I$, then

   $$R/I \xrightarrow{\cdot x} R/I$$

   is the zero-map. But R/I is not the zero-module, so there is some non-zero element which is multiplied by x which is not a zero divisor, so R/I is not torsion-free, and therefore not flat. □

Proposition 4.7 (Characterisation of flatness)**.** *The following are equivalent:*

1. *T_M preserves exactness of all exact sequences.*
2. *T_M preserves exactness of short exact sequences.*
3. *T_M is flat, (i.e. it preserves injectivity)*

4. *If $f : N \to N'$ is R-linear, injective, and N and N' are finitely-generated R-modules, then*

$$\mathrm{Id}_M \otimes f : M \otimes N \to M \otimes N'$$

is injective.

Proof. The easy part of this proof goes (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4). For (1) $\Rightarrow$ (2), short exact sequences are a subset of all exact sequences. For (2) $\Rightarrow$ (3), consider that preserving injectivity is the same as preserving exact sequences like

$$0 \to N \to M,$$

and that we can always "complete" this kind of sequence to a short exact sequence via

$$0 \to N \to M \to M/N \to 0.$$

This is preserved, since it is a short exact sequence, so the "injective" exact sequence is also preserved. And for (3) $\Rightarrow$ (4), clearly flatness implies flatness for finitely generated modules.

Now we need to do the "hard work."

(2) $\Rightarrow$ (1) Assume

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is exact. Then

$$0 \to A/\ker(f) \xrightarrow{\bar{f}} B \xrightarrow{g} \mathrm{Im}(g) \to 0$$

is short-exact (where $\bar{f}$ is the induced map by the quotient). Then we have

$$0 \to M \otimes A/\ker(f) \xrightarrow{\mathrm{Id}_M \otimes \bar{f}} M \otimes B \xrightarrow{\mathrm{Id}_M \otimes g} M \otimes \mathrm{Im}(g) \to 0$$

is exact by hypothesis. So we know that
$\ker(\mathrm{Id}_M \otimes g) = \mathrm{Im}(\mathrm{Id}_M \otimes \bar{f}) = \mathrm{Im}(\mathrm{Id}_M \otimes f)$. Therefore

$$M \otimes A \to M \otimes B \to M \otimes C$$

is exact.

(4) $\Rightarrow$ (3) We won't do the whole proof of this, but we can get the flatness of general modules, since if anything were 0, then there would be a finitely generated submodule, and we would be able to reduce to (4), and then we would get a contradiction. There's more detail in his notes.

(3) $\Rightarrow$ (2) T_M is right exact, flatness means that you preserve the left side, and then since the map preserves the right side as well, we are done.

□

Proposition 4.8. *Let*

$$f : R \to S$$

be a ring homomorphism, and M be a flat R-module. Then $S \otimes_R M$ is a flat S-module.

Proof. Take $g : N \to N'$, an injective S-linear map. Then

$$\begin{array}{ccc} (S \otimes_R M) \otimes_S N & \xrightarrow{\mathrm{Id}_{S\otimes M} \otimes g} & (S \otimes_R M) \otimes_S N' \\ \downarrow & & \downarrow \\ M \otimes N & \xrightarrow{\mathrm{Id}_M \otimes g} & M \otimes N' \end{array}$$

the diagram commutes, and the bottom arrow is thus injective since M is flat, therefore the top arrow must be injective, and so $S \otimes_R M$ is a flat S-module.

4.3 Further Examples of Tensor Products

Lecture 8

Example 4.9. Consider

$$x \otimes y \in \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}.$$

We can also write this as

$$x \otimes y = n \cdot \frac{x}{n} \otimes y = \frac{x}{n} \otimes ny = 0$$

So $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0$ for any n. We used the fact that $\mathbb{Q}$ is a divisible group, i.e.

$$\forall x \in \mathbb{Q},\ \forall n \geqslant 1,\ \exists y \in \mathbb{Q} \,:\, n \cdot y = x,$$

as well as the fact that $\mathbb{Z}/n\mathbb{Z}$ is a torsion group, i.e. all elements are of finite order. In fact, any divisible group tensored with any torsion group must be 0. In particular

$$\mathbb{Q}/\mathbb{Z} \otimes \mathbb{Q}/\mathbb{Z} = 0.$$

But (from the last question of example sheet 1), for any R-module $M \neq 0$, if M finitely generated, then $M \otimes_R M \neq 0$[6]

Example 4.10. Let V be a $\mathbb{Q}$-vector space, then we have

$$\begin{aligned} \mathbb{Q} \otimes_{\mathbb{Q}} V &\xrightarrow{\sim} V \\ x \otimes v &\mapsto xv \end{aligned}$$

[6]Luckily for us $\mathbb{Q}/\mathbb{Z}$ is not finitely generated. Phew, maths has survived another seeming inconsistency.

But also, we have

$$\begin{aligned} \mathbb{Q} \otimes_{\mathbb{Z}} V &\xrightarrow{\sim} V \\ x \otimes v &\mapsto xv \end{aligned}$$

Proof. Every tensor is pure, take

$$\begin{aligned} \sum \frac{a_i}{b_i} \otimes v_i &= \sum \frac{1}{b_i} \otimes a_i v_i \\ &= \sum \frac{1}{b_i} \otimes (b_i \cdot \frac{a_i}{b_i} v_i) \end{aligned}$$

where $a_i, b_i \in \mathbb{Z}$, $v_i \in V$. Then we can rewrite this as

$$\sum 1 \otimes \frac{a_i}{b_i} v_i = 1 \otimes \sum \frac{a_i}{b_i} v_i$$

so every tensor must be pure. Then have

$$1 \otimes v \to v$$

so this map is surjective. And to show injective, it is enough to show it is injective on pure tensors:

$$\begin{aligned} x \otimes v \mapsto xv = 0 \Rightarrow\ & x = 0 \text{ or } v = 0 \\ & \Rightarrow x \otimes v = 0 \end{aligned}$$

Example 4.11.

$$\begin{aligned} M \otimes_R (\bigoplus_i N_i) &\xrightarrow{\sim} \bigoplus_i (M \otimes N_i) \\ m \otimes (n_i)_{i \in I} &\mapsto (m \otimes n_i)_{i \in I}. \end{aligned}$$

We also have a map

$$M \otimes \Pi_{i \in I} N_i \to \Pi_{i \in I} (M \otimes N_i),$$

however, we have not put the isomorphism sign here, because it is not in general an isomorphism. An example where this is not an isomorphism is:

$$\mathbb{Q} \otimes_{\mathbb{Z}} \Pi_{n \geqslant 1} \frac{\mathbb{Z}}{2^n \mathbb{Z}}$$

is not isomorphic to

$$\Pi_{n \geqslant 1} \mathbb{Q} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{2^n \mathbb{Z}}$$

This second expression is zero, since $\mathbb{Q}$ is division, and each $\mathbb{Z}/2^n\mathbb{Z}$ is torsion, so we are tensoring a bunch of 0 modules, and so the overall product is 0.

However, in the first expression, if we take an element of infinite order:

$$g := (1, 1, 1, \dots) \in \Pi_{n \geqslant 1},$$

and no matter how many times you add this element to itself, it will never vanish. Therefore, the order of this element is infinite, so the group that it generates is isomorphic to $\mathbb{Z}$. Then

$$\mathbb{Q} \otimes_{\mathbb{Z}} \langle g \rangle \cong \mathbb{Q}.$$

But, there is an injective map

$$\langle g \rangle \hookrightarrow \Pi_{n \geqslant 1} \mathbb{Z}/2^n\mathbb{Z}$$

Then we can tensor with $\mathbb{Q}$ to get

$$\mathbb{Q} \otimes \langle g \rangle \hookrightarrow \mathbb{Q} \otimes_{\mathbb{Z}} \Pi_{n \geqslant 1} \mathbb{Z}/2^n\mathbb{Z}.$$

The question, then, is why does this preserve injectivity? Well, $\mathbb{Q}$ is a flat $\mathbb{Z}$-module. The proof of this is postponed for now, but we will see it later.

Example 4.12. Let's look at $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ as a $\mathbb{C}$-algebra. First, what is $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ as a $\mathbb{C}$-module. Well, first we identify the second $\mathbb{C}$ with $\mathbb{R}^2$. and then we extend scalars from $\mathbb{R}$ to $\mathbb{C}$ by tensoring the first $\mathbb{C}$.

So we get

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \otimes \mathbb{R}^2 \cong \mathbb{C}^2.$$

Now, let's think about it as a $\mathbb{C}$-algebra. We do the same thing as before, by identifying $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ with $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^2$. To make the algebra more explicit, we do this as:

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \otimes_{\mathbb{R}} \frac{\mathbb{R}[T]}{(T^2 + 1)}$$

But this is congruent to

$$\frac{\mathbb{C}[T]}{(T^2 + 1)} \cong \frac{\mathbb{C}}{(T - i)} \times \frac{\mathbb{C}}{(T + i)} \cong \mathbb{C} \times \mathbb{C}$$

Then if we follow an element through these maps, we get

$$\begin{aligned}
(a + bi) \otimes (c + di) &\mapsto (a + bi) \otimes (c + dT + (T^2 + 1)\mathbb{R}[T]) \\
&\mapsto (a + bi)(c + dT) + (T^2 + 1)\mathbb{C}[T] \\
&\mapsto ac + bdiT + (ibc + Tad) := p \\
p &\mapsto (p + (T - i)\mathbb{C}[T], p + (T + i)\mathbb{C}[T]) \\
&\mapsto ((ac - bd) + i(bc + ad), ac + bd + i(bc - ad)) = (xy, x\bar{y})
\end{aligned}$$

So $x \otimes y \mapsto (xy, x\bar{y})$.

5 Localization

5.1 Localization of Rings and Modules

Definition 5.1 (Multiplicative Set)**.** A **multiplicative subset** S of R is a subset such that

- $1 \in S$
- If $a, b \in S$ then $a, b \in S$.

If $U \subseteq R$ is any set, then the multiplicative closure S of U is the intersection of all multiplicative subsets of R containing U, also definable as

$$\{\Pi_{i=1}^{n} u_i \mid n \geqslant 0,\ u_i \in U\}$$

Examples 5.2.

1. If R is an integral domain, then $S = R \setminus \{0\}$ is multiplicative.
2. If P is a prime ideal of R, then $S = R \setminus P$ is multiplicative.
3. If $x \in R$, then $S = \{x^n \mid n \geqslant 0\}$ is multiplicative.

Example 5.3. $\mathbb{Q}$ is obtained from $\mathbb{Z}$ by "adding inverses" for the elements of the multiplicative subset $\mathbb{Z} \setminus \{0\}$, and we have a ring homomorphism

$$\mathbb{Z} \hookrightarrow \mathbb{Q}$$

Localization wil let us generalize to general R, not just $\mathbb{Q}$, and general multiplicative subsets, but we will generally not have injectivity of the ring homomorphism.

Definition 5.4 (Localization)**.** Let $S \subset R$ be a multiplicative set, and M an R-module, then consider $M \times S$. We define

$$(m_1, s_1) \sim (m_2, s_2)$$

if there exists a u in S such that $u(s_2 m_1 - s_1 m_2) = 0$. This defines an equivalence relation. Reflexivity and symmetry are obvious, and transitivity follows from S being multiplicative.

Then we write

$$\frac{m}{s} := \left[\frac{m}{s}\right]_{\sim}$$

$$S^{-1}M = \left\{\frac{m}{s} \mid (m, s) \in M \times S\right\}.$$

Also have

$$\frac{m_1}{s_1} + \frac{m_2}{s_2} = \frac{s_2 m_1 + s_1 m_2}{s_1 s_2}$$

defines a well-defined abelian group. We define $S^{-1}R$ defined by thinking of R as an R-module, and then we make $S^{-1}R$ into a ring by

$$\frac{m_1}{s_1} \cdot \frac{m_2}{s_2} = \frac{m_1 m_2}{s_1 s_2}.$$

Then we can obtain an $S^{-1}R$-module structure on $S^{-1}M$, in the obvious way.

We also have a localization map

$$\begin{aligned} R &\to S^{-1}R \\ r &\mapsto \frac{r}{1} \end{aligned}$$

and on the module:

$$\begin{aligned} M &\to S^{-1}M \\ m &\mapsto \frac{m}{1} \end{aligned}$$

which are R-linear.

We will show that $\sim$ is an equivalence relation.

Proof. It is clearly reflexive and symmetric, and for transitivity, we have:

$$\begin{aligned} (m_1, s_1) &\sim (m_2, s_2) \\ (m_2, s_2) &\sim (m_3, s_3) \end{aligned}$$

Then there are $u, v \in S$ such that:

$$\begin{aligned} u(s_2 m_1 - s_1 m_2) &= 0 \\ v(s_3 m_2 - s_2 m_3) &= 0 \end{aligned}$$

If we multiply the first thing by $v \cdot s_3$ and the second thing by $u \cdot s_1$, then add them, we cancel to get

$$uv(s_2(s_3 m_1 - s_1 m_3)) = 0$$

and $uvs_1 \in S$ since S is a multiplicative set. □

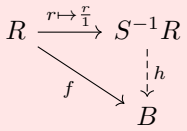
Proposition 5.5 (Universal property of $S^{-1}R$)**.** *If $U \subseteq R$ any subset, and $S \subseteq R$ the multiplicative closure of U. And also*

$$f : R \to B$$

a ring homomorphism to any ring B such that $f(u)$ is a unit for all $u \in U$. Then

$$\exists! h : S^{-1}R \to B$$

such that



commutes. So we have

$$\mathrm{Hom}_{Ring}(S^{-1}R, B) \leftrightarrow \left\{\phi \in \mathrm{Hom}_{Ring}(R, B) \,|\, \phi(U) \subseteq B^\times\right\}.$$

Proof. Take some $f : R \to B$ a ring homomorphism such that $f(U) \subseteq B^\times$. Then immediately, we know $f(S) \subseteq B^\times$. Then we want $h : S^{-1}R \to B$ such that $f(r) = h\left(\frac{r}{1}\right)$, for all $r \in R$.

Then h must satisfy

$$1 = h(1) = h\left(\frac{1}{s} \cdot \frac{s}{1}\right) = h\left(\frac{1}{s}\right) f(s).$$

This means that $h\left(\frac{1}{s}\right) = f(s)^{-1}$. So

$$h\left(\frac{r}{s}\right) = h\left(\frac{1}{s}\right) h\left(\frac{r}{1}\right) = f(s)^{-1} f(r).$$

So we have an h which is defined, but is it *well*-defined. If we have

$$\frac{r_1}{s_1} = \frac{r_2}{s_2}$$

then there is some t such that

$$t(r_1 s_2 - r_2 s_1) = 0.$$

Therefore $ts_2 r_1 = ts_1 r_2$, now we apply f to get

$$f(t)f(s_2)f(r_1) = f(t)f(s_1)f(r_2)$$

Then if we multiply by $f(t)^{-1}f(s_1)^{-1}f(r_2)^{-1}$, we get

$$\frac{f(r_1)}{f(s_1)} = \frac{f(r_2)}{f(s_2)}.$$

So h is well-defined, and then also gives a ring homomorphism. □

Proposition 5.6. *If (A, j) has the same universal property as $(S^{-1}R, \iota_{S^{-1}R})$ where ι maps r to $\frac{r}{1}$, then we have*

$$\begin{aligned} S^{-1}R &\xrightarrow{\sim} A \\ \frac{r}{s} &\mapsto j(s)^{-1}j(r) \end{aligned}$$

Proof. In notes, exact same idea as for the one in tensor products. □

Facts:

1. Take $\frac{r}{s} \in S^{-1}R$, then
$$\frac{r}{s} = \frac{0}{1} \Leftrightarrow \exists u \in S,\, ur = 0.$$
2. So $S^{-1}R = 0 \Leftrightarrow 0 \in S$ (this can be a useful way to analyse multiplicative subsets, i.e. show that the localised ring is the zero ring).
3. $\ker(\iota_{S^{-1}R}) = \{r \in R \,|\, \exists u \in S \text{ s.t. } ur = 0\}$
4. $\iota_{S^{-1}R} : R \to S^{-1}R$ is injective $\Leftrightarrow$ S doesn't contain any zero divisors.
5. ι is always an epimorphism, but usually not surjective.
Example. $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is an epimorphism. If $f, g : \mathbb{Q} \to A$ such that
$$f \circ \iota = g \circ \iota$$
then $f = g$. But ι is clearly not surjective.

Example 5.7.

1. $f \in R$, $S = \{f^n \,|\, n \geqslant 0\}$ Then we write
$$R_f = S^{-1}R.$$
For example, $R = \mathbb{Z}$, $f = 2$. Then
$$R_2 = \left\{ \frac{a}{2^n} \,|\, a \in \mathbb{Z},\, n \geqslant 0 \right\} \cong \mathbb{Z}\left[\frac{1}{2}\right].$$
There is a minor notational problem, we write
$$\mathbb{Z}/n\mathbb{Z} \text{ for the finite ring}$$
and
$$\mathbb{Z}_2 \text{ for the 2-adic integers}$$
in this course. Usually it won't matter because we won't care too much about the p-adics.
2. $p \in \operatorname{spec} R$ (where $\operatorname{spec} R$ is the set of prime ideals of R). Then $S = R \setminus p$ is a multiplicative set, and
$$R_p = (R \setminus p)^{-1}R.$$
Take $R = \mathbb{Z}$ and $p = (3)$, then
$$\mathbb{Z}_{(3)} = \left\{ \frac{a}{b} \,|\, b, a \in \mathbb{Z},\, 3 \nmid b \right\}$$

Proposition 5.8. *Let M be an R-module, $S \subseteq R$ a multiplicative set, then*

$$\begin{aligned} S^{-1}R \otimes_R M &\to S^{-1}M \\ \frac{r}{s} \otimes m &\mapsto \frac{rm}{s} \end{aligned}$$

Proof. We can define a bilinear map

$$\begin{aligned} S^{-1}R \times M &\to S^{-1}M \\ \left(\frac{r}{s}, m\right) &\mapsto \frac{rm}{s} \end{aligned}$$

which is R-bilinear. Therefore, from the universal property of the tensor product, we have an $S^{-1}R$-linear map

$$\begin{aligned} \varphi : S^{-1}R \otimes_R M &\to S^{-1}M \\ \frac{r}{s} \otimes m &\mapsto \frac{rm}{s} \end{aligned}$$

φ is surjective, obviously. ($\frac{1}{s} \otimes m \mapsto \frac{m}{s}$.)

For injectivity, note that every term

$$\sum_i \frac{r_i}{s_i} \otimes m_i \in S^{-1}R \otimes_R M$$

is pure. Let

$$s = \Pi_i s_i$$

and

$$t_j = \prod_{i \neq j} s_i$$

Then

$$\sum_i \frac{r_i}{s_i} \otimes m_i = \sum \frac{1}{s_i} \otimes r_i m_i = \sum \frac{t_i}{s} \otimes r_i m_i$$

Then the t_i can be moved across, since it is an element of R, so we obtain

$$\sum \frac{1}{s} \otimes t_i r_i m_i = \frac{1}{s} \sum t_i r_i m_i$$

Then if

$$\varphi\left(\frac{1}{s} \otimes m\right) = \frac{0}{1},$$

we must have some $u \in S$ such that

$$u(1 \cdot m - 0 \cdot s) = 0,$$

so $um = 0$. Then

$$\frac{1}{s} \otimes m = \frac{u}{us} \otimes m = \frac{1}{us} \otimes um = 0$$

so φ is injective.

Lecture 9

$S^{-1}R \otimes (\,\cdot\,)$ acts on modules and on morphisms. And $S^{-1}(\,\cdot\,)$ acts on modules, but we can show that it acts on morphisms too via the universal property. If $f : N \to N'$ is an R-linear map, we can produce the following commutative diagram

$$\begin{array}{ccc} S^{-1}R \otimes_R N & \xrightarrow{\mathrm{Id}_{S^{-1}R} \otimes f} & S^{-1}R \otimes_R N' \\ \Big\downarrow{\scriptstyle \frac{1}{s}\otimes n \mapsto \frac{n}{s}} & & \Big\downarrow{\scriptstyle \frac{1}{s}\otimes n' \mapsto \frac{n'}{s}} \\ S^{-1}N & \xrightarrow{\quad S^{-1}f \quad} & S^{-1}N' \end{array}$$

which commutes if we define $(S^{-1}f)(\frac{n}{s}) = \frac{f(n)}{s}$, i.e. $S^{-1}f$ lets f act only on the numerator, and not the denominator.

Then, just to spell out what this means, $S^{-1}R \otimes_R (\,\cdot\,) \cong S^{-1}(\,\cdot\,)$ is a natural isomophism (respecting both the objects and the R-linear maps).

Remark. If A is an R-algebra, then

$$S^{-1}R \otimes_R A \simeq S^{-1}A$$

is an R-linear isomorphism between these structures as modules. But this is also an isomorphism between the structures as algebras (just verify that multiplication is preserved).

Lemma 5.9. *Let M be an $S^{-1}R$-module. Then define $S^{-1}M$ to be the module obtained by restricting scalars from $S^{-1}R$ to R on M, and then apply $S^{-1}M$. Then*

$$\begin{aligned} S^{-1}M &\cong M \\ \frac{m}{s} &\mapsto \frac{1}{s}m \end{aligned}$$

as $S^{-1}R$-modules. Equivalently

$$\begin{aligned} M &\cong S^{-1}R \otimes_R M \\ m &\mapsto \frac{1}{1} \otimes m \end{aligned}$$

as $S^{-1}R$-modules.

Proof. We have

$$M \to S^{-1}M$$
$$m \mapsto \frac{m}{1}$$

is $S^{-1}R$-linear. This is surjective, since $\frac{1}{s}m \mapsto \frac{m}{s}$. Injectve, since $\frac{m}{1} = \frac{0}{1}$ means that there exists a $u \in S$ such that $um = 0$ So $m = 0$. □

Recall: Last time we saw there is a bijection:

$$\mathrm{Hom}_{\mathrm{Rings}}(S^{-1}R, B) \overset{\sim}{\leftrightarrow} \{f \in \mathrm{Hom}_{\mathrm{Rings}}(R, B) \mid f(S) \subseteq B^{\times}\}$$

Now, we are going to show that for M a fixed R-module, and L any $S^{-1}R$-module

$$\mathrm{Hom}_R(M, L) \overset{\sim}{\leftrightarrow} \mathrm{Hom}_R(S^{-1}M, L).$$

Now let's state this formally:

Proposition 5.10. *Let M be an R-module, L an $S^{-1}R$-module, and*

$$f : M \to L$$

is an R-linear map. Then:

$$\exists! h : S^{-1}M \to L$$

which is $S^{-1}R$-linear, such that $f = h \circ \iota_{s^{-1}M}$. I.e. $f(m) = h(\frac{m}{1})$.

Proof. We know that

$$S^{-1}(\,\cdot\,) \cong S^{-1}R \otimes_R (\,\cdot\,)$$

Then we have ι defined as

$$\iota : M \to S^{-1}R \otimes_R M$$
$$m \mapsto \frac{1}{1} \otimes m$$

So now take $f : M \to L$ which is R-linear, and to begin we seek

$$h : S^{-1}R \otimes_R M \to S^{-1}R \otimes_R L$$

. So let

$$h = \mathrm{Id}_{S^{-1}R} \times f.$$

However, we want the range to be L, not $S^{-1}R \otimes_R L$. But luckily we have the previous lemma which tells us that "unnecessary localisation" is just like doing nothing. So write

$$h\left(\frac{r}{s} \otimes m\right) = \frac{r}{s} f(m)$$

So $h(1 \otimes m) = f(m)$. And the uniqueness of h comes from the fact that $\{1 \otimes m\}_{m \in M}$ generate $S^{-1}R \otimes M$ as an $S^{-1}R$-module, so once we're defined on these elements, we're defined on all of $S^{-1}R \otimes_R M$. □

Theorem 5.11 ($S^{-1}(\,\cdot\,)$ is exact)**.** *If*

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is an exact sequence of R-modules, then

$$S^{-1}A \xrightarrow{S^{-1}f} S^{-1}B \xrightarrow{S^{-1}g} C$$

is an exact sequence of $S^{-1}R$ modules. And therefore, since it preserves all exact sequences of length 3, it preserves all exact sequences of any length.

Proof. First, we will show that the image is contained in the kernel.

$$(S^{-1}g) \circ (S^{-1}f) = S^{-1}(g \circ f) = 0$$

So $\operatorname{Im}(S^{-1}f) \subseteq \ker(S^{-1}g)$.

Now we need to show that the kernel is contained in the image. So take $\frac{b}{s} \in \ker(S^{-1}g)$, then $\frac{g(b)}{s} = \frac{0}{1}$. Therefore, we have

$$\exists u \in S.\, u \cdot g(b) = 0,$$

but then, since g is R-linear, we have

$$g(ub) = 0.$$

SO $ub \in \ker(g) = \operatorname{Im}(f)$ So

$$\exists a.\, f(a) = ub.$$

Then

$$\frac{b}{s} = \frac{ub}{us} = \frac{f(a)}{us} = (S^{-1}f)\left(\frac{a}{us}\right).$$

Therefore $\frac{b}{s} \in \operatorname{Im}(S^{-1}f)$. Equivalently, $S^{-1}R$ is flat R-module □

This gives a proof that $\mathbb{Q}$ is a flat $\mathbb{Z}$-module, since $\mathbb{Q}$ is a field of fractions, and hence a localisation of $\mathbb{Z}$. Now, since $S^{-1}(\,\cdot\,)$ is exact, if $N \leqslant M$ as R-modules, then

$$\begin{gathered} N \xrightarrow{\iota} M \\ S^{-1}N \xrightarrow{S^{-1}\iota} S^{-1}M \\ \frac{n}{s} \mapsto \frac{n}{s} \end{gathered}$$

Proposition 5.12. *Let M be an R-module, and N, P be submodules of M. Then*

1. $S^{-1}(N+P) = S^{-1}N + S^{-1}P$
2. $S^{-1}(N \cap P) = S^{-1}N \cap S^{-1}P$
3. $S^{-1}M/S^{-1}N \xrightarrow{\sim} S^{-1}(M/N)$ *via*

$$\frac{m}{s} + S^{-1}N \mapsto \frac{m+N}{s}$$

Proof.

1. There is not much to say, we have

$$\frac{n+p}{s} = \frac{n}{s} + \frac{p}{s} \in S^1N + S^{-1}P,$$

and we have

$$\frac{n}{s_1} + \frac{p}{s_2} = \frac{s_2n + s_1p}{s_1s_2} \in S^{-1}(N+P).$$

2. $\subseteq$ is clear.

For $\supseteq$, take $x \in S^{-1}N \cap S^{-1}P$, then

$$x = \frac{n}{s_1} = \frac{p}{s_2}.$$

This means that there must exist some $u \in S$ such that:

$$us_0n = us_1p = w$$

Then $us_0n \in N$ and $us_1p \in P$, so $w \in N \cap P$. Now, take the element

$$x = \frac{n}{s_1} = \frac{us_2n}{us_1s_2} = \frac{w}{us_1s_2}.$$

Then $w \in N \cap P$ and the denominator is in S, so $x \in S^{-1}(N \cap P)$.

3. $0 \longrightarrow N \xrightarrow{\iota} M \xrightarrow{\pi} M/N \longrightarrow 0$

is exact, so

$$0 \longrightarrow S^{-1}N \xrightarrow{S^{-1}\iota} S^{-1}M \xrightarrow{S^{-1}\pi} S^{-1}(M/N) \longrightarrow 0$$

$(S^{-1}\iota)(S^{-1}N) = S^{-1}N$. Then $S^{-1}\pi\left(\frac{m}{s}\right) = \frac{m+N}{s}$. So we have the required isomorphism, it comes basically immediately from exactness.

Proposition 5.13 (I)**.** *f M, N are R-modules, then:*

$$S^{-1}M \otimes_{S^{-1}R} S^{-1}N \cong S^{-1}(M \otimes_R N)$$

Proof. Need to show that:

$$(S^{-1}R \otimes_R M) \otimes_{S^{-1}R} (S^{-1}R \otimes_R N) \cong S^{-1}R \otimes_R (M \otimes_R N).$$

We already know that this statement is true from our discussion of extension of scalars for the tensor product. □

We have the special case where, if $\mathfrak{p}$ is a prime ideal of R, then

$$M_{\mathfrak{p}} \otimes_{R_{\mathfrak{p}}} N_{\mathfrak{p}} \cong (M \otimes N)_{\mathfrak{p}}.$$

5.2 Extension and contraction of ideals along $R \to S^{-1}R$

If $f : A \to B$ is a ring homomorphism, then if we take the preimage of an ideal $\mathfrak{b}$ of B, then $f^{-1}(\mathfrak{b})$ is called a contracted ideal of A. And if we take an ideal $\mathfrak{a}$ of A, then $(f(\mathfrak{a}))$ is an extended ideal of B. We already saw on example sheet 1 that

$$\{\text{ contracted ideals of } A\} \leftrightarrow \{\text{ extended ideals of } B\}.$$

Now, we will look at this in the special case of the localization map:

$$R \to S^{-1}R$$
$$r \mapsto \frac{r}{1}$$

We have the **extension map** from ideals of R to ideals of $S^{-1}R$

$$\mathfrak{a} \mapsto \mathfrak{a}^e = S^{-1}\mathfrak{a}$$

and the **contraction map**:

$$\mathfrak{b} \mapsto \mathfrak{b}^c$$

We claim that:

$$\mathfrak{a}^e = \left\{ \frac{a}{s} \mid a \in \mathfrak{a},\, s \in S \right\}.$$

Lecture 10 To prove this,we realise that $\mathfrak{a}^e$ is generated by $\left\{\frac{a}{1} \mid a \in \mathfrak{a}\right\}$, and these are obtained under the map.

Then also, we claim:

$$\mathfrak{a}^{ec} = \cup_{s \in S}(\mathfrak{a} : s),$$

where $(I : x) = \{r \in R : rs \in \mathfrak{a}\}$.

Proof. Take $r \in \cup_{s\in S}(\mathfrak{a} : s)$, so $rs = a$ in R, so in $S^{-1}R$, we have:

$$\frac{rs}{1} = \frac{a}{1} \Longrightarrow \frac{r}{1} = \frac{a}{s}$$

Therefore $r \in \mathfrak{a}^{ec}$, since the elements of $\mathfrak{a}^e$ are those of the form $\frac{a}{s}$, and so r gets sent to this element under the extension, so is in the contraction of the extension.

Now assume that $r \in \mathfrak{a}^{ec}$, then

$$\begin{aligned}
&\Rightarrow \frac{r}{1} = \frac{a}{s} \\
&\Rightarrow \exists u \in S,\ \ u(rs - a) = 0 \\
&\Rightarrow rus = ua \\
&\Rightarrow r \in (\mathfrak{a} : us)
\end{aligned}$$

and $us \in S$. □

Now, to consider contraction, if $\mathfrak{b}$ is an ideal of $S^{-1}R$, then
$\mathfrak{b}^c = \left\{r \in R \,|\, \frac{r}{1} \in \mathfrak{b}\right\}$.

Claim: $\mathfrak{b}^{ce} = \mathfrak{b}$.

Proof. We've proved on example sheet 1 that $\subseteq$ always holds.

To prove the reverse inclusion, take $\frac{r}{s} \in \mathfrak{b}$ (for some $r \in R, s \in S$). Then $\frac{r}{1} \in \mathfrak{b}$ by multiplying by $\frac{s}{1}$. So $r \in \mathfrak{b}^c$, and so $\frac{r}{1} \in \mathfrak{b}^{ce}$, and so $\frac{r}{s} \in \mathfrak{b}^{ce}$ since, again, ideals are multiplicatively closed.

Proposition 5.14. *Consider*

$$\begin{aligned}
R &\to S^{-1}R \\
r &\mapsto \frac{r}{1}
\end{aligned}$$

Then

1. *Every ideal of $S^{-1}R$ is extended.*
2. *An ideal $\mathfrak{a}$ of R is contracted if and only if the image of S in $R/\mathfrak{a}$ contains no zero divisors of $R/\mathfrak{a}$.*
3. *$\mathfrak{a}^e = R$ if and only if $\mathfrak{a} \cap S \neq \emptyset$.*
4. *We have a bijection:*

$$\begin{aligned}
\{\mathfrak{p} \in \operatorname{spec} R \,|\, \mathfrak{p} \cap S = \emptyset\} &\leftrightarrow \operatorname{spec} S^{-1}R \\
\mathfrak{p} &\mapsto \mathfrak{p}^e \\
\mathfrak{q}^c &\hookleftarrow \mathfrak{q}
\end{aligned}$$

Proof.

1. True since $\mathfrak{b}^{ce} = \mathfrak{b}$ for all ideals $\mathfrak{b}$ of $S^{-1}R$.

2. One inclusion was proved on example sheet 1, so we have that $\mathfrak{a}$ is contracted $\Leftrightarrow \mathfrak{a}^{ec} = \cup_{s\in S}(\mathfrak{a} : s) \subseteq \mathfrak{a}$. So

$$\mathfrak{a} \text{ is contracted } \Leftrightarrow \forall r \in R(Sr \cap \mathfrak{a} \neq \emptyset \Rightarrow r \in \mathfrak{a}) \qquad (*)$$

   Since if $Sr \cap \mathfrak{a} \neq \emptyset$ but $r \notin \mathfrak{a}$, then we have an element in this union which is not contained in $\mathfrak{a}$.

   But $r \in \mathfrak{a}$ means that $r + \mathfrak{a} = 0 + \mathfrak{a} \in R/\mathfrak{a}$, and $Sr \cap \mathfrak{a} \neq \emptyset$ means that $0 + \mathfrak{a} \in S \cdot (r + \mathfrak{a})$. So (*) is true is true if and only if the image $\bar{S}$ of S in $R/\mathfrak{a}$ contains no zero divisors. (Because if it does contain a zero divisor s, then $s \cdot r_0 \in \mathfrak{a}$ in R, for some $r_0 \notin \mathfrak{a}$).

3. Assume that $\mathfrak{a} \cap S \neq \emptyset$, then we can take some element $x \in \mathfrak{a} \cap S$, and

$$\frac{x}{x} \in \mathfrak{a}^e$$

   since $x \in \mathfrak{a}$, and $x \in S$. But

$$\frac{x}{x} = 1,$$

   so $\mathfrak{a}^e = S^{-1}R$.

   Conversely, if $\frac{1}{1} \in \mathfrak{a}^e = R$, then $\frac{1}{1} = \frac{a}{s}$ for some $a \in \mathfrak{a}$, some $s \in S$. Therefore $\exists u \in S$, $u(s - a) = 0 \Rightarrow us = ua \in S \cap \mathfrak{a}$.

4. Look at the following contraction map:

$$\{\mathfrak{p} \in \operatorname{spec} R \,|\, \mathfrak{p} \cap S = \emptyset\} \leftarrow \operatorname{spec} S^{-1}R$$

   Then we have:

   (a) A contraction of a prime ideal is always prime.

   (b) $\mathfrak{p} \subseteq R$ is contracted $\Leftrightarrow$ the image of $\bar{S}$ in $R/\mathfrak{p}$ contains no zero divisors. But then, $R/\mathfrak{p}$ is an integral domain, so this is equivalent to $0 \notin \bar{S}$, i.e. $S \cap \mathfrak{p} = \emptyset$.

   So we have shown that $\{\mathfrak{p} \in \operatorname{spec} R \,|\, \mathfrak{p} \cap S = \emptyset\}$ is the set of contracted prime ideals.

   Finally, this map is injective, since:

$$\mathfrak{q}^c \leftarrow\!\shortmid \mathfrak{q}$$

   and then $\mathfrak{q}^{ce} = q$, since this is true for any ideal in the localisation map.

Now we need to show the other direction, for $\mathfrak{p} \in \operatorname{spec} R$ such that $\mathfrak{p} \cap S = \emptyset$, we have shown that $\mathfrak{p}$ is contracted, so $\mathfrak{p}^{ec} = \mathfrak{p}$. But we still need to show that such a $\mathfrak{p}$ (which is disjoint from S) is in $\operatorname{spec} S^{-1}R$, i.e. is a prime ideal.

Want: $S^{-1}R/\mathfrak{p}^e$ is a domain. We know that $\mathfrak{p}^e \neq S^{-1}R$, by (3) above, and the fact that $\mathfrak{p} \cap S = \emptyset$. So we just need to show that $S^{-1}R/\mathfrak{p}^e$ has no zero divisors. To do this, we will embed $S^{-1}R/\mathfrak{p}^e$ in $\operatorname{Frac}(R/\mathfrak{p})$:

Proof. Consider the map

$$R \twoheadrightarrow R/\mathfrak{p} \hookrightarrow \operatorname{Frac} R/\mathfrak{p}$$

has the property that all elements of S are sent to units (since $S \cap \mathfrak{p} = \emptyset$). Then by the universal property of $S^{-1}R$, we have a map

$$\begin{aligned} \varphi : S^{-1}R &\to \operatorname{Frac} R/\mathfrak{p} \\ \frac{r}{s} &\mapsto \frac{r+\mathfrak{p}}{s+\mathfrak{p}}. \end{aligned}$$

It suffices to show that $\ker \varphi = \mathfrak{p}^e$. Take $\frac{r}{s} \in \ker \varphi$, then

$$\frac{r+\mathfrak{p}}{s+\mathfrak{p}} = \frac{0}{1}.$$

(in $\operatorname{Frac}(R/\mathfrak{p})$.) Now $\operatorname{Im} \varphi \subseteq \bar{S}^{-1}(R/\mathfrak{p})$ where $\bar{S}$ is the image of S in $R/\mathfrak{p}$. Then we can restrict φ to get

$$\begin{aligned} \varphi : S^{-1}R &\to \bar{S}^{-1}(R/\mathfrak{p}) \\ \frac{r}{s} &\mapsto \frac{r+\mathfrak{p}}{s+\mathfrak{p}} \end{aligned}$$

So $\varphi\left(\frac{r}{s}\right) = \frac{0}{1} \Rightarrow \exists u + \mathfrak{p} \in \bar{S}$ such that

$$\begin{aligned} (u+\mathfrak{p})(r+\mathfrak{p}) &= 0 \\ ur + \mathfrak{p} &= 0 \end{aligned}$$

Therefore $ur \in \mathfrak{p}$, and $u \in S$. Therefore, we can write

$$\frac{r}{s} = \frac{ur}{us} \in \mathfrak{p}^e$$

since $ur \in \mathfrak{p}$ and $us \in S$.

For the other direction, take $x \in \mathfrak{p}^e$, then $x = \frac{p}{s}$ for $p \in \mathfrak{p}$, and $s \in S$. Then

$$\varphi(x) = \frac{p+\mathfrak{p}}{s+\mathfrak{p}} = 0.$$

Therefore $x \in \ker \varphi$.

□

Application: If I is an ideal of R, then the radical of I:

$$\sqrt{I} = \{r \in R \mid \exists n \geqslant 1,\, r^n \in I\}$$

($I \subseteq \sqrt{I}$).

Proposition 5.15.

$$\sqrt{I} = \bigcap_{I \subseteq \mathfrak{p} \in \operatorname{spec} R} \mathfrak{p}$$

Proof. Take $x \in \sqrt{I}$. Then $x^n \in I$, so for every $\mathfrak{p} \in \operatorname{spec} R$, if $I \subseteq \mathfrak{p}$, then $x^n \in \mathfrak{p}$, so $x \in \mathfrak{p}$.

Conversely, take $x \in R$, $x \notin \sqrt{I}$, so $I \neq R$, and therefore $R/I \neq 0$. Let $\bar{x}$ be the image of x in R/I. Then consider

$$(R/I)_{\bar{x}},$$

the localisation of R/I at $\bar{x}$ i.e. $\{\bar{x}^n | n \geqslant 1\}^{-1}(R/I)$. This is *not* the zero ring, because we did not invert 0, because $x \notin \sqrt{I}$ by assumption. Therefore $(R/I)_{\bar{x}}$ has a prime ideal, because it has a maximal ideal. This prime ideal corresponds to a prime ideal of R/I which does not contain $\bar{x}$, and *this* prime ideal corresponds to a prime ideal of R which contains I and does not contain x.

5.3 Local Properties

Lecture 11

Definition 5.16 (Local ring)**.** A ring R is **local** if

$$|\operatorname{mspec} R| = 1.$$

That is, the set of all maximal ideals of R contains only one object.

Then we can also write $(R, \mathfrak{m})$, when we want to give a name to the unique maximal ideal of R.

Example 5.17. We obtain local rings by taking any ring and localising at a prime ideal. Let $\mathfrak{p} \in \operatorname{spec} R$, then there is a bijection

$$\begin{aligned} \{\mathfrak{n} \in \operatorname{spec} R \mid \mathfrak{n} \subseteq \mathfrak{p}\} &\overset{\sim}{\leftrightarrow} \operatorname{spec} R_{\mathfrak{p}} \\ \mathfrak{n} &\mapsto \mathfrak{n}^e \\ \mathfrak{q}^c &\mapsfrom \mathfrak{q} \end{aligned}$$

so, we can see that all prime ideals of $R_{\mathfrak{p}}$ are contained in $\mathfrak{p}R_{\mathfrak{p}}$. So $(R_{\mathfrak{p}}, \mathfrak{p}R_{\mathfrak{p}})$ is a local ring.

Example 5.18. Recall that

$$\mathbb{Z}_{(2)} = \{\frac{a}{b} \,|\, a, b \in \mathbb{Z},\, 2 \nmid b\}.$$

Then this ring is local with unique maximal ideal:

$$(2)\mathbb{Z}_{(2)} = \{\frac{2a}{b} \,|\, a, b \in \mathbb{Z},\, 2 \nmid b\}$$

Proposition 5.19. *For M an R-module, the following are equivalent:*

1. $M = 0$.
2. $M_{\mathfrak{p}} = 0\ \forall \mathfrak{p} \in \operatorname{spec} R$.
3. $M_{\mathfrak{m}} = 0\ \forall \mathfrak{m} \in \operatorname{mspec} R$.

This is a "local property".

Proof. 1. $\Rightarrow$ 2. $\Rightarrow$ 3. is clear.

Now, assume 3, and that $M \neq 0$. Take $0 \neq m_0 \in M$. Consider the annihilator $\operatorname{Ann}_R(m_0) = \{r \in R \,|\, rm_0 = 0\}$, which is an ideal of R. It is a proper ideal (because it does not contain 1, since $m_0 \neq 0$).

We want to show that some localisation of M is nonzero at some maximal ideal to get a contradiction, so take $\mathfrak{m} \in \operatorname{mspec} R$ such that $\operatorname{Ann}_R(m_0) \subseteq \mathfrak{m}$. Then $\frac{m_0}{1} \in M_{\mathfrak{m}} = 0$, since every element in $M_{\mathfrak{m}}$ is 0 because we assumed 3. Then this means explicitly that:

$$\frac{m_0}{1} = \frac{0}{1} \in M_{\mathfrak{m}}.$$

This means, therefore that $um_0 = 0$ for some $u \in R \setminus \mathfrak{m}$ (since this is the multiplicative set). But then an element not in $\mathfrak{m}$ is *not* in the annihilator, but we've shown that this element annihilates m_0. Contradiction. □

Proposition 5.20 (T)**.** *FAE:*

1. *f is injective.*
2. *$f_{\mathfrak{p}} : M_{\mathfrak{p}} \to N_{\mathfrak{p}}$ is injective $\forall \mathfrak{p} \in \operatorname{spec} R$.*
3. *$f_{\mathfrak{m}} : M_{\mathfrak{m}} \to N_{\mathfrak{m}}$ is injective $\forall \mathfrak{m} \in \operatorname{mspec} R$.*

Similarly for surjective instead of injective.

Proof. First, recall that $f_{\mathfrak{p}}\left(\frac{m}{s}\right) = \frac{f(m)}{s}$.

(1) ⇒ (2) Localising at $\mathfrak{p}$ is an exact functor (so it preserves injectivity and surjectivity).

(2) ⇒ (3) Every maximal ideal is prime.

(3) ⇒ (1) We assume 3, and have the following exact sequence

$$0 \to \ker f \xrightarrow{\iota} M \xrightarrow{f} N.$$

So now we apply localisation and get

$$0 \to (\ker f)_{\mathfrak{m}} \to M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}}} N_{\mathfrak{m}} \tag{*}$$

is exact, since localisation is an exact functor. But (*) is exact precisely means that $\ker(f_{\mathfrak{m}}) = (\ker f)_{\mathfrak{m}}$. But $\ker(f_{\mathfrak{m}}) = 0$ by assumption of 3. So we have proved that $(\ker f)_{\mathfrak{m}} = 0$ for every maximal ideal, so the kernel itself must be 0, because of our previous proposition (being 0 is a local property).

Proposition 5.21 (L)**.** *et M be an R-module, then TFAE:*

1. *M is a flat R-module*
2. *$M_{\mathfrak{p}}$ is a flat $R_{\mathfrak{p}}$ module $\forall \mathfrak{p} \in \operatorname{spec} R$*
3. *$M_{\mathfrak{m}}$ is a flat $R_{\mathfrak{m}}$ module $\forall \mathfrak{m} \in \operatorname{mspec} R$.*

Proof.

(1) ⇒ (2) We know that flatness is localisable, since $M_{\mathfrak{p}} \cong R_{\mathfrak{p}} \otimes M$ is an $R_{\mathfrak{p}}$-linear isomorphism, and this is an extension of scalars.

(2) ⇒ (3) Since every maximal ideal is prime.

(3) ⇒ (1) Take $f : N \to P$, this is R-linear and injective, now take some maximal ideal $\mathfrak{m} \in \operatorname{mspec} R$, then $f_{\mathfrak{m}} : N_{\mathfrak{m}} \to P_{\mathfrak{m}}$ is injective, by the proposition above. But then, we also have that

$$N_{\mathfrak{m}} \otimes_{R_{\mathfrak{m}}} M_{\mathfrak{m}} \xrightarrow{f_{\mathfrak{m}} \otimes \mathrm{Id}_{M_{\mathfrak{m}}}} P_{\mathfrak{m}} \otimes M_{\mathfrak{m}}$$

is injective, by 3. But then, localizing and then tensoring, is the same as tensoring and then localizing (which follows from extension of scalars). So we have a commuting diagram:

$$\begin{array}{ccc} N_{\mathfrak{m}} \otimes_{R_{\mathfrak{m}}} M_{\mathfrak{m}} & \xrightarrow{f_m \otimes \mathrm{Id}_{M_{\mathfrak{m}}}} & P_{\mathfrak{m}} \otimes_{R_{\mathfrak{m}}} M_{\mathfrak{m}} \\ \downarrow \sim & & \downarrow \sim \\ (N \otimes_R M)_{\mathfrak{m}} & \xrightarrow{(f \otimes \mathrm{Id}_M)_{\mathfrak{m}}} & (P \otimes_R M)_{\mathfrak{m}} \end{array}$$

since this commutes, $(f \otimes \mathrm{Id}_M)_{\mathfrak{m}}$ is injective for all $\mathfrak{m} \in \operatorname{mspec} R$. So by the previous proposition $(f \otimes \mathrm{Id}_M)$ is injective, so M is flat. □

Definition 5.22 (Locally free)**.** An R-module M is **locally free** if $M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$-module $\forall \mathfrak{p} \in \operatorname{spec} R$.

Example 5.23. This property is not localizable, take $R = \mathbb{C} \times \mathbb{C}$. Then

$$\operatorname{spec} R = \{\mathbb{C} \times \{0\}, \{0\} \times \mathbb{C}\}$$

If we have a map

$$\begin{aligned} \mathbb{C} \times \mathbb{C} &\to \mathbb{C} \\ (a, b) &\mapsto b \end{aligned}$$

This sends $\mathbb{C} \times \mathbb{C} \setminus \mathbb{C} \times \{0\}$ to units. Therefore we can localise by the universal property, to get

$$\begin{aligned} (\mathbb{C} \times \mathbb{C})_{\mathbb{C} \times \{0\}} &\to \mathbb{C} \\ \frac{(a, b)}{(c, d)} &\mapsto \frac{b}{d} \end{aligned}$$

which is clearly surjective since $\frac{(0,b)}{(0,1)} \mapsto b$, but also this is injective (very clear since b must be 0, and then check that the overall tuple is 0 in the localization, but it's also in notes). So $(\mathbb{C} \times \mathbb{C})_{\mathbb{C} \times \{0\}} \cong \mathbb{C}$ and $(\mathbb{C} \times \mathbb{C})_{\{0\} \times \mathbb{C}} \cong \mathbb{C}$. So for every $\mathbb{C} \times \mathbb{C}$-module M and every $\mathfrak{p} \in \operatorname{spec} \mathbb{C} \times \mathbb{C}$, $M_{\mathfrak{p}}$ is a $\mathbb{C}$-module, so it's a vector space, and therefore it must be free. *But* not every module over $\mathbb{C} \times \mathbb{C}$ is free, for example, take $M = \mathbb{C} \times \{0\}$ as a $\mathbb{C} \times \mathbb{C}$-module, now prove that M is not free of rank 0 (not the zero module), and also that it is not free of rank $\geqslant 1$ (any element is linearly dependent by itself, so cannot be part of a basis). So M is locally-free, but not free.

5.4 The localization of a ring as a quotient

Let $U \subseteq R$ be a subset, and let $S \subseteq R$ be the multiplicative closure of U. Now define

$$R_U := \frac{R[\{T_u\}_{u \in U}]}{(\{u \cdot T_u - 1\}_{u \in U})}$$

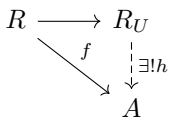
Then the claim is that $R_U \cong S^{-1}R$ as rings, also as R-algebras. The isomorphism is given by:

$$T_u \mapsto \frac{1}{u},$$

and in the other direction

$$r \cdot T_{u_1} \cdots \cdot T_{u_l} + I_U \leftarrow\!\shortmid \frac{r}{u_1 \cdots u_l}$$

Proof. The idea is to show that R_U has the universal property of $S^{-1}R$:



where A is any ring and f is any ring homomorphism. Then A is an R-algebra via f, and the diagram commutes if and onlt if h is an R-algebra homomorphism.

$$\mathrm{Hom}_{R-\mathrm{alg}}(R_U, A) \cong \{\varphi : U \to A \mid f(u) \cdot h(u) = 1\}.$$

Special case: R localised at a single element x for some $x \in R$ is isomorphic to $R[T]/(xT - 1)$.

6 Nakayama's Lemma

Proposition 6.1 (Cayley-Hamilton)**.** *Let M be a finitely generated R-module, and $f : M \to M$ an R-linear map. $\mathfrak{a} \subseteq R$ an ideal of R.*

If $f(M) \subseteq \mathfrak{a}M$, then there exists a_i such that

$$f^n + a_1 f^{n-1} + \cdots + a_n f^0 = 0$$

where $a_i \in \mathfrak{a}$. (f^n means f applied n times.)

Write $M = \mathrm{span}_R\{m_1, \ldots, m_n\}$. Then $\mathfrak{a}M = \mathrm{span}_{\mathfrak{a}}\{m_1, \ldots, m_n\}$. So we can write

$$\begin{pmatrix} f(m_1) \\ \vdots \\ f(m_n) \end{pmatrix} = P \cdot \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix}$$

where P is an $n \times n$ matrix with coefficients in $\mathfrak{a}$. If we take $\rho : R \to \mathrm{End}\, M$ to be the structure ring homomorphism of M as an R-module. Then have

$$\begin{aligned} R[T] &\to \mathrm{End}(M) \\ T &\mapsto f \end{aligned}$$

a map compatible with ρ. Then

$$T \cdot \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = P \cdot \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix},$$

or by rearranging:

$$Q \cdot \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = 0,$$

where $Q = T \cdot I_n - P$. Then we can multiply by the adjugate of Q to get

$$(\det Q) \cdot \begin{pmatrix} m_1 \\ \vdots \\ m_n \end{pmatrix} = 0$$

So $(\det Q) \cdot m = 0$ for every $m \in M$, since these elements generate M. This means that $m \mapsto (\det Q) \cdot m$ is the zero of the endomorphism ring. But $m \mapsto (\det Q) \cdot m$, is the function which takes $\det Q$ (which is a polynomial in T) and substitutes $T = f$, and moreover $\det Q$ is a monic polynomial with coefficients in $\mathfrak{a}$. □

Corollary 6.2. *Let M be a finitely generated R-module, and $\mathfrak{a} \subseteq R$ an ideal.*

If $\mathfrak{a}M = M$, then there is some $a \in \mathfrak{a}$ such that $am = m$ $\forall m \in M$.

Proof. Apply Cayley-Hamilton with $f = \mathrm{Id}_M$. All the conditions are met, and Cayley-Hamilton gives us $(1 + a_1 + \cdots + a_n) \cdot \mathrm{Id}_M = 0$. Take $a = -a_1 - a_2 - \cdots - a_n$, and we are done. □

Definition 6.3 (Jacobson Radical)**.** The **Jacobson radical** $J(R)$ is given by:

$$J(R) = \cap_{m \in \mathrm{mspec}\, R} \mathfrak{m}$$

Example 6.4.

1. $(R, \mathfrak{m})$, a local ring, has $J(R) = \mathfrak{m}$.
2. $J(\mathbb{Z}) = \{0\}$.

Proposition 6.5. *For $x \in R$, we have*

$$x \in J(R) \Leftrightarrow 1 - xy \text{ is a unit of } R,\ \forall y \in R$$

Proof. Assume $x \in J(R)$ and assume (for contradiction) that $1 - xy$ is not a unit for some $y \in R$. Then if $1 - xy$ is not a unit, then $R \neq (1 - xy)$, so there is a maximal ideal containing $(1 - xy)$. We know $x \in J(R)$, so x is in every maximal ideal, therefore $1 - xy \in \mathfrak{m}$, and $xy \in \mathfrak{m}$. So $1 - xy + xy \in \mathfrak{m}$.

For the converse, assume $x \notin J(R)$, then there is a $\mathfrak{m}$ such that $x \notin \mathfrak{m}$. Then $\mathfrak{m} + (x)$ is the entire ring. Therefore, we have $t \in \mathfrak{m}$ and $y \in R$, such that $t + xy = 1$, or equivalently $1 - xy = t$ for $t \in \mathfrak{m}$. So $1 - xy$ cannot be a unit, otherwise $\mathfrak{m}$ would contain a unit and be R. □

Lemma 6.6 (Nakayama's Lemma)**.** *If M is a finitely generated R-module, and $\mathfrak{a} \subseteq J(R)$ is an ideal of R such that $\mathfrak{a}M = M$. Then $M = 0$.*

Proof. By the corollary we had, there is $a \in \mathfrak{a}$ such that $am = m \ \forall m \in M$. So $(1 - a)m = 0$. But we know by the above proposition that $1 - a$ is a unit of R. Therefore $m = 0$ for any $m \in M$. □

Corollary 6.7. *Let M be a finitely generated R-module, and $N \subseteq M$ an R-submodule, and $\mathfrak{a} \subseteq J(R)$ an ideal.*

If $N + \mathfrak{a}M = M$, then $N = M$.

Proof.

$$\mathfrak{a} \cdot (M/N) = \mathfrak{a}M/N = (\mathfrak{a}M + N)/N = M/N.$$

Then by Nakayama $M/N = 0$ so $N = M$. □

7 Integral and finite extensions Part I

Definition 7.1 (Integral)**.** Let A be an R-algebra, and $x \in A$. x is **integral** over R (or R-integral) if

$$\exists f \in R[T],\ f \text{ monic s.t. } f(x) = 0.$$

Example 7.2.

1. K a field, A a K-algebra, $X \in A$. X is K-integral iff X is K-algebraic.
2. We will show later that:
    1. The $\mathbb{Z}$-integral elements of $\mathbb{Q}$ is just $\mathbb{Z}$.
    2. The $\mathbb{Z}$-integral elements of $\mathbb{Q}[\sqrt{2}]$ are $\mathbb{Z}[\sqrt{2}]$.
    3. The $\mathbb{Z}$-integral elements of $\mathbb{Q}[\sqrt{5}]$ is $\mathbb{Z}\left[\frac{1+\sqrt{5}}{2}\right]$ [7].

[7] "Sometimes it's what you expect, and sometimes it's what you expect if you have just a *bit* more experience."

Definition 7.3 (Faithful module)**.** An R-module M is **faithful** if the structure ring homomorphism from R to the ring of endomorphisms of M is injective.

That is to say, for every $0 \neq r \in R$, there is an $m \in M$ such that $rm \neq 0$.

Example 7.4. $R \subseteq A$ rings (A an R-algebra), so A is an R-module in the natural way. And in fact, A *must* be faithful, because if $r \neq 0 \in R$, then $r \cdot 1_A = r \neq 0$.

Proposition 7.5. *$R \subseteq A$ are rings, $x \in A$, then $R[x] \subseteq A$. This means that A is an $R[x]$ module. Then x is R-integral if and only if there exists $M \subseteq A$ such that*

1. *M is a faithful $R[x]$-module*
2. *M is f.g. as an R-module.*

The first condition means that M is an R-submodule of A, $xM \subseteq M$, and M is faithful over $R[x]$.

Proof. First, assume that conditions 1 and 2 hold. Then $xM \subseteq M$, and we have an R-linear map

$$\begin{aligned} f : M &\to M \\ m &\mapsto xm \end{aligned}$$

M is a finitely generated R-module, so by Cayley-Hamilton, we have that

$$f^n + r_1 f^{n-1} + \cdots + r_n f^0 = 0$$

in $\mathrm{End}_R(M)$. Now, evaluate this at some $m \in M$, we obtain that

$$(x^n + r_1 x^{n-1} + \cdots + r_n x^0)m = 0 \ \ \forall m \in M.$$

Then, since M is a faithful $R[x]$-module, we conclude that $x^n + r_1 x^{n-1} + \cdots + r^0 = 0$, and so x is integral over R.

For the other direction, assume x is R-integral, then

$$x^n + r_1 x^{n-1} + \cdots + r_n x^0 = 0$$

for $r_i \in R$. Then

$$M = \mathrm{span}_R\{x^0, \ldots, x^{n-1}\}$$

satisfies $xM \subseteq M$ (since $x \cdot x^{n-1} = -r_1 x^{n-1} - \cdots - r_n x^0$). So M is clearly finitely generated, and is faithful over $R[x]$ since $1_A = x^0 \in M$. □

Lecture 12

Definition 7.6 (Integral/Finite algebra)**.** Let A be an R-algebra, then

1. A is **integral** over R if every element of A is R-integral.
2. A is **finite over** R if A is finitely generated as an R-module.

Proposition 7.7. *Let A be an R-algebra, then TFAE:*

1. *A is a finitely generated integral R-algebra*
2. *A is generated as an R-algebra by a finite set of integral elements.*
3. *A is finite over R.*

Proof.

(1) $\Rightarrow$ (2) Trivial. Generated by a finite set and all elements are integral.

(2) $\Rightarrow$ (3) A is generated by $\alpha_1, \ldots, \alpha_m$ (all R-integral) as an R-algebra. Then α_i integral means that

$$\alpha_i^{n_i} + r_{i,1}\alpha_i^{n_i-1} + \cdots + r_{i,n_i}\alpha_i^0 = 0.$$

So $\alpha_i^{n_i} \in \operatorname{span}_R\{\alpha_i^0, \ldots, \alpha_i^{n_i-1}\}$. Then for all $e_1, \ldots, e_n \geqslant 0$ such that $\alpha_1^{e_1} \ldots \alpha_m^{e_m} \in \operatorname{span}_R\{\alpha_1^{f_1}, \ldots \alpha_m^{f_m}\}$ where $0 \leqslant f_i \leqslant n_i - 1$. This is a finite set which generates A as an R-module. (We do this by reducing the powers in the monomial in the left until we get something which is in the span of the right hand side.)

(3) $\Rightarrow$ (1) A is f.g. as an R-module $\Rightarrow$ A is f.g. as an R-algebra.

Now, to prove that it is integral over R, take $\alpha \in A$, and we want to prove that α is R-integral. Let $\rho : R \to A$ be the structure ring homomorphism of A as an R-algebra. Then $\rho(R) \subseteq A$, and $(\rho(R))[\alpha] \subseteq A$. A is a $\rho(R)[\alpha]$-module, but also must be faithful, because $1_A \subseteq A$. Also A is a finitely-generated $\rho(R)[\alpha]$-module. So by proposition 7.5, we must have α is $\rho(R)$-integral. So α is R-integral.

□

Proposition 7.8. *A is an R-algebra, $O = \{a \in A | a$ is R-integral$\}$, then O is an R-subalgebra of A.*

Proof. Take $x, y \in O$. Then $\{x, y\}$ is a finite set of R-integral elements, so it must generate an integral R-subalgebra of A. $x + y$, $x \cdot y$, $x - y$ are all, therefore, in O. Also O contains the image of R, since $(T - r)|_{T=r\cdot 1_A} = 0$ so $r \cdot 1_A \in O$. □

Proposition 7.9.

1. *If C is finite over B, B is finite over A, then C is finite over A.*
2. *If C is integral over B, B is integral over A, then C is integral over A.*

Proof.

1. $C = \text{span}_B\{\gamma_1, \ldots, \gamma_n\}$. $B = \text{span}_A\{\beta_1, \ldots, \beta_l\}$, then easy to see that C is the A-span of all the products $\gamma_i\beta_j$.
2. Let $c \in C$. We want to show that c is A-integral. We know that c is B-integral. So $f(c) = 0$ for $f = T^n + b_1T^{n-1} + \cdots + b_nT^0 \in B[T]$. Then $f \in (A[b_1, \ldots, b_n])[T]$. Call $A[b_1, \ldots, b_n] := A'$. Then $A \subseteq A' \subseteq A'[c]$, and both of these extensions are generated by finitely many integral elements (over A and then over A' respectively). This tells us that both extensions are finite. Then by 1., $A \subseteq A'[c]$ is finite. Then $A \subseteq A'[c]$ is integral, so c is A-integral. So integrality is transitive.

□

Definition 7.10 (Integral Closure)**.**

1. $A \subseteq B$ rings.
   (a) The **integral closure** of A in B is $\bar{A} = \{b \in B \mid b \text{ is A-integral.}\}$
   (b) A is integrally closed in B if $\bar{A} = A$.
2. A is an integral domain. Then
   (a) The **integral closure** of A (with no other ring defined) is the integral closure of A in $\text{Frac}\, A$.
   (b) A is **integrally closed** if A is integrally closed in $\text{Frac}\, A$.

Example 7.11. 1. $\mathbb{Z}[\sqrt{5}]$ is not integrally closed, since

$$\alpha = \frac{1+\sqrt{5}}{2} \in \mathbb{Q}[\sqrt{5}] \setminus \mathbb{Z}[\sqrt{5}].$$

We have $\alpha^2 - \alpha - 1 = 0$, so α *is* $\mathbb{Z}[\sqrt{5}]$-integral.

2. $\mathbb{Z}$ and $k[T_1, \ldots, T_n]$ are integrally closed.

Proposition 7.12. *If A is a UFD then A is integrally closed.*

Proof. Let $X \in \mathrm{Frac}(A) \setminus A$. Then $X = \frac{a}{b}$ where $a \in A$, $b \in A$. Reduce until there is some $p \mid b$ but $p \nmid a$ for some p a prime element of A. If x is A-integral, then we can write:

$$\left(\frac{a}{b}\right)^n + a_1\left(\frac{a}{b}\right) + \cdots + a_n\left(\frac{a}{b}\right) = 0$$

Then we can mutiply through by b^n to get

$$a^n = -b(a_1 b^1 a^{n-1} + +a_2 b^2 a^{n-2} + \cdots + a_n b^n)$$

But then $p \mid b$, so $p \mid a^n$ so $p \mid a$. Contradiction. □

Lemma 7.13. *If A and B are rings, and $\bar{A}$ is the integral closure of A in B, then $\bar{A}$ is integrally closed in B.*

Proof. This follows from the transitivity of integrality. If $x \in B$ is integral over $\bar{A}$, then $A \subseteq \bar{A} \subseteq \bar{A}[x]$. Each of these extensions is integral. So $A \subseteq \bar{A}[x]$ must be integral. So x is A integral, and so $x \in \bar{A}$.

Proposition 7.14. *If $A \subseteq B$ are rings, then*

1. *If B integral over A, then*
   - *$B/\mathfrak{b}$ is integral over $A/A \cap \mathfrak{b}$ for all ideals $\mathfrak{b}$ of B.*
   - *$S^{-1}B$ is integral over $S^{-1}A$ for all multiplicative sets $S \subseteq A$.*
2. *If $\bar{A}$ is the integral closure of A in B, then $S^{-1}\bar{A}$ is the integral closure of $S^{-1}A$ in $S^{-1}B$ (i.e. $\overline{S^{-1}A} = S^{-1}\bar{A}$).*

Proof. In notes.

Lemma 7.15. *$A \subseteq B$ an integral extension of rings, then*

1. *$A \cap B^\times = A^\times$ (units of A).*
2. *If A, B are domains then A a field $\Leftrightarrow$ B a field.*

Proof.

1. $\supseteq$ is clear.

   $\subseteq$: Take $a \in A \cap B^\times$, then $ab = 1$ for some $b \in B$. We want to show $b \in A$. b is integral over A. Therefore $b^n + a_1 b^{n-1} + \cdots + a_n b^0 = 0$ where $a_i \in A$. Then we multiply by a^{n-1} to get:

   $$b + a_1 + a \cdot a_2 + a^2 \cdot a_3 + \cdots + a^{n-1} \cdot a_n = 0.$$

Then

$$b = -(a_1 + a \cdot a_2 + a^2 \cdot a_3 + \cdots + a^{n-1} \cdot a_n)$$

so $b \in A$.

2. Assume B is a field. Then $A^\times = A \cap B^\times = A \cap (B \setminus \{0\}) = A \setminus \{0\}$. Therefore A is a field.

   Now assume that A is a field. Take a non-zero element $0 \neq b \in B$, we want to show that $b \in B^\times$. Well b is A-integral, so

   $$b^n + a_1 b^{n-1} + \cdots + a_n b^0 = 0$$

   for coefficients in A. Let $n \geqslant 1$ be minimal such that this holds. Then write this as

   $$b(b^{n-1} + a_1 b^{n-1} + \cdots + a_{n-1} b^0) = -a_n$$

   Then $b \neq 0$ and $b^{n-1} + a_1 b_{n-2} + \cdots + a_{n-1} b^0 \neq 0$ by the minimality of n. Therefore $a_n \neq 0$ since B is a domain, but then a_n is a unit, since A is a field. Therefore

   $$b(b^{n-1} + a_1 b_{n-2} + \cdots + a_{n-1} b^0)(-a_n^{-1}) = 1.$$

   So b is a unit.

Corollary 7.16. *Let $A \subseteq B$ be an integral extension of rings, and $\mathfrak{q} \in \operatorname{spec} B$. Then*

$$\mathfrak{q} \in \operatorname{mspec} B \Leftrightarrow \mathfrak{q} \cap A \in \operatorname{mspec} A$$

Proof. There is a ring embedding

$$A/\mathfrak{q} \cap A \hookrightarrow B/\mathfrak{q}.$$

These are both integral domains, and $B/\mathfrak{q}$ is an integral extension of $A/\mathfrak{q} \cap A$. Therefore by the previous theorem, $A/\mathfrak{q} \cap A$ is a field $\Leftrightarrow$ $B/\mathfrak{q}$ is a field. □

8 Noether normalization and Hilbert's Nullstellensatz

8.1 Noether normalization

Definition 8.1 (k-algebraically independent)**.** If A is a k-algebra, $X_1, \dots X_n \in A$. Then $X_1, \dots, X_n$ are **k-algebraically independent** if

$$\forall 0 \neq p \in k[T_1, \dots, T_n], \;\; p(X_1, \dots, X_n) \neq 0.$$

I.e. the k-algebra homomorphism

$$\begin{aligned} k[T_1, \dots T_n] &\to A \\ T_i &\mapsto X_i \end{aligned}$$

is injective.

Theorem 8.2 (Noether Normalization)**.** *If $A \neq 0$ is a finitely geneated k-algebra, then*

$$\exists x_1, \dots, x_n \in A \quad n \geqslant 0$$

which are k-algebraically independent, such that:

$$A \text{ is finite over } k[X_1, \dots, X_n]$$

Example. (Proof method)**.** If $A = k[T, T^{-1}]$.

First, $k[T] \subseteq k[T, T^{-1}]$ is not a finite extension.

Proof. Assume that it were a finite extension. Then T^{-1} is integral over $k[T]$. But then the integrality relation says that

$$(T^{-1})^n \in \text{span}_{k[T]}\{(T^{-1})^0, \dots, (T^{-1})^{n-1}\}$$

But then if we multiply by T^n, we get that

$$1 \in \text{span}_{k[T]}\{T^n, T^{n-1}, \dots, T^1\}$$

which is obviously not true. So we get a contradiction. □

But take $c \in k$ a scalar which we will choose later. Then

$$A = k[T, T^{-1}] = k[T, T^{-1} - cT]$$

So we claim that $k[T^{-1} - cT] \subseteq A$ is a finite extension for most values of $c \in k$.

Proof. We have

$$\begin{aligned} T \cdot T^{-1} - 1 = 0 \\ \Rightarrow ((T^{-1} - cT) + cT) \cdot T - 1 = 0 \\ \Rightarrow c \cdot T^2 + (T^{-1} - cT) \cdot T - 1 = 0 \end{aligned}$$

And $c \in k$, $(T^{-1} - cT) \in k[T^{-1} - cT]$ and $-1 \in k[T^{-1} - cT]$. So as long as $c \neq 0$, we can multiply through by c^{-1}, so then T is integral over $k[T^{-1} - cT]$.

Proof of Noether Normalization (8.2). We will prove it assuming that k is infinite[8], but the proof will work in general.

Our strategy will be to induct on the minimal number m of generators of A as a k-algebra.

Base case: If $m = 0$ then $A = k$. So take the "polynomial algebra" over 0 variables, which is just k itself.

Inductive step: Assume that A is generated by $X_1, \ldots, X_m \in A$. If $X_1, \ldots, X_m$ are algebraically independent, then take the polynomial algebra just to be A itself.

Otherwise, we take claim (*) to be: $\exists c_1, \ldots c_{m-1} \in k$ such that X_m is integral over $B = k[X_1 - c_1X_m, X_2 - c_2X_m, \ldots, X_{m-1} - c_{m-1}X_m]$.

First we prove that claim (*) is sufficient:

$$A = B[X_m] \Rightarrow A \text{ finite over } B$$

since X_m is integral by (*). Now, by the inductive hypothesis, we have $n < m$ and $Z_1, \ldots, Z_n \in B$ which are k-algebraically independent, and such that B is finite over $k[Z_1, \ldots, Z_n] := A'$. So A is finite over A' by transitivity of finiteness. Therefore (*) is sufficient

Proof of (*): We have by assumption that $X_1, \ldots, X_m \in A$ are *not* algebraically independent over k. Therefore, $\exists 0 \neq f \in k[T_1, \ldots, T_n]$ such that $f(X_1, \ldots, X_m) = 0$.

Want: X_m is integral over $k[X_1 - c_1X_m, X_2 - c_2X_m, \ldots, X_{m-1} - c_{m-1}X_{m-1}]$.

Write f as the sum of its homogeneous parts,

(for example $f = (T_1^2T_2 + 3T_2^3) + (2T_1T_3 + T_2T_3) + (T_2 + T_3) + 1$, where we collected terms of like degree.)

[8] for pedagogical reasons.

Let $F \neq 0$ be the part of degree $\deg f$ (i.e. of highest degree). Then for $c_1, \ldots, c_{m-1} \in k$ which we will choose later, we have:

$$f(T_1 + c_1 T_m, \ldots, T_{m-1} + c_{m-1} T_{m-1}, T_m) := g(T_1, \ldots, T_m)$$

becomes equal to

$$F(c_1, \ldots, c_{m-1}, 1) \cdot T_m^r + \begin{cases} \text{terms of degree } < r \text{ in } T_m \\ \text{with coefficients in } k[T_1, \ldots, T_{m-1}] \end{cases}$$

(this is very analogous to what we did in the example after the statement of Noether Normalization (8.2)). Now notice

1. $g(X_1 - c_1 X_m, \ldots, X_{m-1} - c_{m-1} X_m, X_m) = f(X_1, \ldots, X_m) = 0$
2. g as a polynomial in T_m over $k[T_1, \ldots, T_{m-1}]$ has degree $\leqslant r$, the coefficient of T_m^r is $F(c_1, \ldots, c_{m-1}, 1)$.

Now $F(T_1, \ldots, T_m)$ is a non-zero homogeneous polynomial, so $F(T_1, \ldots, T_{m-1}, 1)$ is not the zero polynomial (see example sheet 1). Therefore, there are $c_1, \ldots, c_{m-1}$ such that $F(c_1, \ldots, c_{m-1}, 1) \neq 0$. This is now an integrality relation, since the highest coefficient is a scalar, so we can divide by it. □

8.2 Hilbert's Nullstellensatz

Proposition 8.3 (Zariski's Lemma)**.** *Let $k \subseteq L$, L a field which is f.g. as a k-algebra. Then* $\dim_k L < \infty$.

Proof. By Noether, we have some

$$k[x_1, \ldots, x_l] \subseteq L$$

such that L is a finite extension. Then finite $\Rightarrow$ integral, so this is an integral extension of integral domains. But then if one of these is a field, then the other must be, and we know that L is a field. So we must have that $k[X_1, \ldots, X_l]$ is a field. Therefore, in fact, this must be a polynomial algebra over 0 variables. So $k \subseteq L$ is finite. □

From now, let $k \subseteq \Omega$ be fields, where Ω is algebraically closed.

Definition 8.4 (Algebraic set)**.**

1. Let S be a subset of $k[T_1, \ldots, T_n]$. Then

$$V(S) = \{\mathbf{x} \in \Omega^n \,:\, f(\mathbf{x}) = 0 \ \forall f \in S\}.$$

   A set of the form $V(S)$ is called a **k-algebraic subset** of Ω^n (or, an algebraic subset of Ω^n defined over k).

2. Let X be a subset of Ω^n. Then

$$I(X) = \{f \in k[T_1, \ldots, T_n] \,:\, f(\mathbf{x}) = 0 \ \forall \mathbf{x} \in X\}.$$

Clearly, if $\mathfrak{a}$ is the ideal of $k[T_1, \ldots, T_n]$ generated by S, then $V(\mathfrak{a}) = V(S)$

Recall that for all $k \subseteq L$ a finite field extension, there is a k-algebra embedding

$$L \to \Omega.$$

Theorem 8.5. *Let $\mathfrak{a} \subseteq k[T_1, \ldots, T_n]$ be an idea, then we have:*

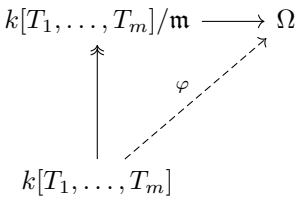
1. ***Weak Nullstellensatz:*** $V(\mathfrak{a}) = \emptyset \Leftrightarrow 1 \in \mathfrak{a}$
2. ***Strong Nullstellensatz:*** $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

Proof.

1. Proving $\Leftarrow$ is clear. For $\Rightarrow$, assume that $1 \notin \mathfrak{a}$. Then $\exists \mathfrak{m} \in \operatorname{mspec} k[T_1, \ldots, T_n]$ such that $\mathfrak{a} \subseteq \mathfrak{m}$.

   This implies that $\frac{k[T_1,\ldots,T_n]}{\mathfrak{m}}$ is a field which is finitely generated over k as an algebra. Therefore $\dim_k \left(\frac{k[T_1,\ldots,T_n]}{\mathfrak{m}}\right) < \infty$.

   But then, there is an injective k-algebra homomorphism:



   where $\ker \varphi = \mathfrak{m}$. Write $\bar{X} = (\varphi(T_1), \ldots, \varphi(T_n)) \in \Omega^n$. For $f \in k[T_1, \ldots, T_n]$ we have $\varphi(f) = f(\bar{X})$. So $\forall f \in \mathfrak{a} \subseteq \mathfrak{m} = \ker \varphi$, we have

$$f(\bar{X}) = \varphi(f) = 0.$$

   And so, $\bar{X} \in V(\mathfrak{a})$.

2. Let $f \in \sqrt{\mathfrak{a}}$, so that $f^l \in \mathfrak{a}$ for some $l \geqslant 1$. Then

$$f^l(\bar{X}) = 0 \ \ \forall \bar{X} \in V(\mathfrak{a})$$

Therefore $f(\bar{X}) = 0$, so $f \in I(V(\mathfrak{a}))$.

Conversely, take $f \in I(V(\mathfrak{a}))$. We want $f \in \sqrt{\mathfrak{a}}$, so we want that $\bar{f}$[9] is nilpotent in $k[T_1, \ldots, T_n]/\mathfrak{a}$, as this would imply that $f \in \sqrt{\mathfrak{a}}$. But this in turn is implied by

$$(k[T_1, \ldots, T_n]/\mathfrak{a})_{\bar{f}} = 0.$$

This is isomorphic to

$$k[T_1, \ldots, T_n, T_{n+1}]/(\mathfrak{a}^e + (T_{n+1}f - 1))$$

Write $\mathfrak{b} = (\mathfrak{a}^e + (T_{n+1}f - 1))$. We want that $1 \in \mathfrak{b}$ to get that the ring is 0. But by the weak Nullstellensatz, this means we want $V(\mathfrak{b}) = \emptyset$.

Take $\bar{X} = (X_1, \ldots, X_{n+1}) \in V(\mathfrak{b}) \subseteq \Omega^{n+1}$. Then write $\bar{X}_0 = (X_1, \ldots, X_n) \in V(\mathfrak{a})$. Then, since $f \in I(V(\mathfrak{a}))$, we have $f(\bar{X}_0) = 0$ so $(T_{n+1}f)(\bar{X}) = 0$. But then $(T_{n+1}f - 1)(\bar{X}) = -1 \neq 0$, which contradicts that $\bar{X} \in V(\mathfrak{b})$. Therefore $1 \in \mathfrak{b}$ by the Weak Nullstellensatz, so the quotient ring is in fact 0, and therefore $\bar{f}$ is nilpotent, which means that $f \in \sqrt{\mathfrak{a}}$.

□

Recall that $\sqrt{\sqrt{I}} = \sqrt{I}$. Now some facts from algebraic geometry:

1. $X \subseteq Y \subseteq \Omega^n \Rightarrow I(X) \supseteq I(Y)$.

   Similarly, $S \subseteq T \subseteq k[T_1, \ldots, T_n] \Rightarrow V(T) \supseteq V(S)$.

2. $S \subseteq k[T_1, \ldots T_n] \Rightarrow S \subseteq I(V(S))$.

3. $X \subseteq \Omega^n \Rightarrow X \subseteq V(I(X))$.

4. If $X \subseteq \Omega^n$ is an algebraic set, then $X = V(I(X))$.

   Proof. If X algebraic, then $X = V(\mathfrak{a})$ for some $\mathfrak{a}$. This implies that $V(I(X)) = V(I(V(\mathfrak{a}))) \subseteq V(\mathfrak{a}) = X$. □

   $X \subseteq \Omega^n \Rightarrow I(X)$ is a radical ideal.

$$I(X) = \sqrt{(}I(X)).$$

[9] the image of f in the quotient ring.

Proposition 8.6. *Let $k = \Omega$ be an algebraically closed field, and $n \geqslant 0$. There is a bijection:*

$$\begin{aligned} \{ \text{ algebraic subsets of } \Omega^n \} &\leftrightarrow \{ \text{ Radical ideals of } \Omega[T_1, \ldots, T_n] \} \\ X &\mapsto I(X) \\ V(\mathfrak{a}) &\hookleftarrow \mathfrak{a} \end{aligned}$$

Proof. Since $I(X)$ is radical, $X = V(I(X))$ this is already done. For the other direction, let $\mathfrak{a} \subseteq k[T_1, \ldots, T_n]$ be a radical ideal. Then $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}} = \mathfrak{a}$ by the strong nullstellensatz. □

Corollary 8.7. *Let Ω be an algebraically closed field. Then there is a bijection*

$$\begin{aligned} \Omega^n &\leftrightarrow \operatorname{mspec} \Omega[T_1, \ldots, T_n] \\ (X_1, \ldots, X_n) &\mapsto (T_1 - X_1, \ldots, T_n - X_n) \end{aligned}$$

Proof. $\mathfrak{m}_{\bar{X}}$ is a maximal ideal[10], since it is the kernel of

$$\begin{aligned} k[T_1, \ldots, T_n] &\twoheadrightarrow \Omega \\ T_i &\mapsto X_i \end{aligned}$$

So we wish to prove that $\mathfrak{m}_{\bar{X}} = I(\{\bar{X}\})$. Well $\mathfrak{m}_{\bar{X}} \subseteq I(\{\bar{X}\}) \subsetneq \Omega[T_1, \ldots, T_n]$. Therefore $\mathfrak{m}_{\bar{X}} = I(\{\bar{X}\})$. $V(\mathfrak{m}_{\bar{X}}) = \{\bar{X}\}$. The claim follows from the inclusion reversing bijection from before. □

Claim: Every prime ideal $\mathfrak{p}$ is radical. Easily seen, since $x^n \in frakp \Rightarrow x \in \mathfrak{p}$.

Definition 8.8 (Irreducible variety)**.** $X \subseteq \Omega^n$ is **irreducibel** if X is not the union of two proper algebraic subsets.

Proposition 8.9. *If $X \subseteq \Omega^n$ is an algebraic set, then X is irreducible iff $I(X)$ is prime.*

Proof. In the notes.

[10] where $\bar{X}$ is a tuple of elements in Ω.

9 Integral and finite extensions - Part II

Definition 9.1 (Integral elements, Integral closure of an ideal)**.** Let $A \subseteq B$ be rings, and $\mathfrak{a} \subseteq A$ an ideal. Then

1. $x \in B$ is **$\mathfrak{a}$-integral** if:

$$\exists f(T) = T^n + a_1 T^{n-1} + \cdots + a_n T^0 \in A[T] \text{ s.t. } f(x) = 0$$

2. The **integral closue** of $\mathfrak{a}$ in B is

$$\{x \in B : x \text{ is } \mathfrak{a}\text{-integral}\}$$

Proposition 9.2. *Let $A \subseteq B$ be rings, and let $\bar{A}$ be the integral closure of A in B. Let $\mathfrak{a} \subseteq A$ be an ideal of A. Then the integral closure of $\mathfrak{a}$ in B is $\sqrt{\mathfrak{a}\bar{A}}$.*

Proof. If $b \in B$ is $\mathfrak{a}$-integral, then

$$b^n + a_1 b^{n-1} + \cdots + a_n b^0 = 0 \tag{$*$}$$

Therefore, $b \in \bar{A}$. Therefore $b^0, \ldots, b^{n-1} \in \bar{A}$, and so, by $(*)$, $b^n \in \mathfrak{a}\bar{A}$. Therefore $b \in \sqrt{\mathfrak{a}\bar{A}}$.

Conversely, assume $b \in \sqrt{\mathfrak{a}\bar{A}}$. Then $b^n \in \mathfrak{a}\bar{A}$ for some $n \geqslant 1$. Then we have

$$b^n = \sum_{i=1}^{m} a_i \cdot X_i \quad m \geqslant 0 \tag{$**$}$$

where $a_i \in \mathfrak{a}$, and $X_i \in \bar{A}$. Then write $M := A[X_1, \ldots, X_m]$, so that M is a finite A-algebra. Then by $(**)$, we have $b^n M \subseteq \mathfrak{a}M$ (as an inclusion of A-modules). Then define

$$f : M \to M$$
$$f(m) = b^n m$$

Then $f(M) \subseteq \mathfrak{a}M$, and f is A-linear. Then by Cayley-Hamilton, we have:

$$f^l + \alpha_1 f^{l-1} + \cdots + \alpha^l f^0 = 0$$

in $\operatorname{End} M$. So

$$(b^n)^l + \alpha_1 (b^n)^{l-1} + \cdots + \alpha^l (b^n)^0 = 0$$

in B. Therefore b is $\mathfrak{a}$ integral. □

Corollary 9.3. *Let $A \subseteq B$ as rings. $\mathfrak{a} \subseteq A$, and $b \in B$. Then*

$$b \text{ is } \mathfrak{a} \text{ integral} \Leftrightarrow b \text{ is } \sqrt{\mathfrak{a}}\text{-integral}$$

Proof. We want $\sqrt{\mathfrak{a}\bar{A}} = \sqrt{\sqrt{a}\bar{A}}$.

$\subseteq$: Clear.

$\supseteq$: $\sqrt{I}^e \subseteq \sqrt{I^e}$. Therefore $\sqrt{\mathfrak{a}}\bar{A} \subseteq \sqrt{\mathfrak{a}\bar{A}}$. Then take radicals of both sides to get:

$$\sqrt{\sqrt{\mathfrak{a}}\bar{A}} \subseteq \sqrt{\mathfrak{a}\bar{A}}$$

□

Proposition 9.4. *If A is an integrally closed integral domain with an ideal $\mathfrak{a}$, and $A \subseteq B$, where B is another integral domain. Then if $x \in B$. Consider* $\operatorname{Frac} A \subseteq \operatorname{Frac} B$. *The following are equivalent:*

1. *b is integral over $\mathfrak{a}$*
2. *b is algebraic over* $\operatorname{Frac} A$ *with minimal polynomial over* $\operatorname{Frac} A$ *of the form*
$$T^n + a_1 T^{n-1} + \cdots + a_n T^0$$
where $a_i \in \sqrt{\mathfrak{a}}$.

Proof. The second claim implies that b is $\sqrt{\mathfrak{a}}$ integral, which implies by corollary 9.3 that b is $\mathfrak{a}$-integral. So $2 \Rightarrow 1$ is done.

Now assume 1, this implies:

$$b^n + a_1 b^{n-1} + \cdots + a_n b^0 = 0$$

Then $h = T^n + a_1 T^{n-1} + \cdots + a_n T^0 \in (\operatorname{Frac} A)[T]$, and $h(b) = 0$. Let $f \in (\operatorname{Frac} A)[T]$ be the minimal polynomial of b over $\operatorname{Frac} A$. Then fix $\operatorname{Frac} A \subseteq \Omega$ where Ω is an algebraically closed field, so that

$$f = \prod_{i=1}^{l} (T - \alpha_i) \tag{$*$}$$

where $\alpha_i \in \Omega$, W.L.O.G. $\alpha_1 = b$. Then we want to show that the coefficients of f are in $\sqrt{\mathfrak{a}}$.

By proposition 9.2 as well as the fact that A is integrally closed, we have that the integral closure of $\mathfrak{a}$ in $\operatorname{Frac} A$ is $\sqrt{\mathfrak{a}}$[11]. Then we need that the coefficients are in $\operatorname{Frac} A$ and are $\mathfrak{a}$-integral. The coefficients must be in $\operatorname{Frac} A$ by definition, and we obtain $\mathfrak{a}$-integrality by expanding the brackets in $(*)$, and seeing that the coefficients of f are sums of products of $\alpha_1, \dots, \alpha_l$. By the proposition, the integral closure of $\mathfrak{a}$ in Ω is closed under sums and products. So we just need to show that each α_i is $\mathfrak{a}$-integral, but α_i and b have the same minimal polynomial f over $\operatorname{Frac} A$. Therefore

$$\exists \varphi_i : (\operatorname{Frac} A)[b] \to (\operatorname{Frac} A)[\alpha_i]$$

where each φ_i is a $\operatorname{Frac}(A)$-algebra isomorphism which takes

$$b \mapsto \alpha_i.$$

So we have

$$h(\alpha_i) = h(\varphi_i(b)) = \varphi_i(h(b)) = 0.$$

□

10 Cohen-Seidenberg Theorems (Going up/down)

Remark. If $\iota : A \hookrightarrow B$, then we can define

$$\begin{aligned} \iota^* &: \operatorname{spec} B \to \operatorname{spec} A \\ \iota^*(\mathfrak{q}) &= \mathfrak{q} \cap A \end{aligned}$$

Proposition 10.1 (Incomparability)**.** *Let $A \subseteq B$ be an integral extension, and $\mathfrak{q}, \mathfrak{q}' \in \operatorname{spec} B$ with $\mathfrak{q} \subseteq \mathfrak{q}'$, but $\mathfrak{q} \cap A = \mathfrak{q}' \cap A$. Then $\mathfrak{q} = \mathfrak{q}'$.*

Proof. Let $\mathfrak{p} = \mathfrak{q} \cap A = \mathfrak{q}' \cap A$. Then define $S = A \setminus \mathfrak{p}$. Then $\mathfrak{q}$ and $\mathfrak{q}'$ are prime ideals of B not intersecting S, therefore $\mathfrak{q} = (S^{-1}\mathfrak{q})^c$. and $\mathfrak{q}' = (S^{-1}\mathfrak{q}')^c$. Then $B_\mathfrak{p} = (A \setminus \mathfrak{p})^{-1}B$[12]. Then we want $\mathfrak{q}B_\mathfrak{p} = \mathfrak{q}'B_\mathfrak{p}$. This can be shown by

$$\mathfrak{q}B_\mathfrak{p} \cap A_\mathfrak{p} = S^{-1}(\mathfrak{q} \cap A) = S^{-1}\mathfrak{p} = \mathfrak{p}A_\mathfrak{p},$$

and similarly for $\mathfrak{q}'B_\mathfrak{p} \cap A_\mathfrak{p} = \mathfrak{p}A_\mathfrak{p}$. $\mathfrak{p}A_\mathfrak{p}$ is a maximal ideal of $A_\mathfrak{p}$. Then $A \subseteq B$ an integral extension implies that

$$A_\mathfrak{p} \hookrightarrow B_\mathfrak{p}$$

is also an integral extension, so $\mathfrak{q}B_\mathfrak{p} \subseteq \mathfrak{q}'B_\mathfrak{p}$ must also be maximal ideals of $B_\mathfrak{p}$. Therefore $\mathfrak{q}B_\mathfrak{p} = \mathfrak{q}'B_\mathfrak{p}$. □

[11] since $\bar{A} = A$.

[12] Since $\{\text{prime ideals of } R \text{ disjoint from } S\} \leftrightarrow \operatorname{spec} S^{-1}R$

Proposition 10.2 (Lying over)**.** *Let $A \subseteq B$ be an integral extension of rings, and $\mathfrak{p} \in \operatorname{spec} A$ a prime ideal. Then*

$$\exists \mathfrak{q} \in \operatorname{spec} B \text{ s.t. } \mathfrak{q} \cap A = \mathfrak{p}$$

(i.e. $i^* : \operatorname{spec} B \to \operatorname{spec} A$ is surjective for $i : A \to B$.)

Proof.
$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \beta \\ A_{\mathfrak{p}} & \longrightarrow & B_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1} B \end{array}$$

Take $\mathfrak{m} \in \operatorname{mspec} B_{\mathfrak{p}}$ maximal, then we can contract $\mathfrak{m}$ back to A in two ways via this commutative diagram.

Let's first contract to $A_{\mathfrak{p}}$. We know that $A_{\mathfrak{p}} \subseteq B_{\mathfrak{p}}$ is an integral extension, since a localization of an integral extension stays an integral extension (7.14). Then $\mathfrak{m} \cap A_{\mathfrak{p}} \in \operatorname{mspec} A_{\mathfrak{p}}$ by (7.16). But $\operatorname{mspec} A_{\mathfrak{p}}$ contains only one element, $\mathfrak{p}A_{\mathfrak{p}}$. And then $\mathfrak{p}A_{\mathfrak{p}}$ contracts to $\mathfrak{p}$ in A under the localization map. Therefore $\mathfrak{m}$ contracts to $\mathfrak{p}$ under the map $A \to A_{\mathfrak{p}} \to B_{\mathfrak{p}}$. But this is the same map as $A \to B \to B_{\mathfrak{p}}$. Therefore, $\beta^{-1}(\mathfrak{m}) \cap A = \mathfrak{p}$. And $\beta^{-1}(\mathfrak{m})$ is a contraction of a maximal ideal, so it is a prime ideal. □

Proposition 10.3 (Going up)**.** *Let $A \subseteq B$ an integral extension of rings, and $\mathfrak{q}_1 \in \operatorname{spec} B$, $\mathfrak{p}_1, \mathfrak{p}_2 \in \operatorname{spec} A$ such that:*

$$\mathfrak{q}_1 \cap A = \mathfrak{p}_1 \qquad\qquad \mathfrak{p}_1 \subseteq \mathfrak{p}_2$$

Then there is a $\mathfrak{q}_2 \supseteq \mathfrak{q}_1$ which lies in $\operatorname{spec} B$ such that $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$.

Proof. We know $\mathfrak{p}_1 = \mathfrak{q}_1 \cap A$, so

$$\begin{aligned} A/\mathfrak{p}_1 &\hookrightarrow B/\mathfrak{p}_2 \\ a + \mathfrak{p}_1 &\to a + \mathfrak{q}_1 \end{aligned}$$

From lying over, we know that there is a $\mathfrak{q}_2/\mathfrak{q}_1 \in \operatorname{spec} B/\mathfrak{q}_1$ where $\mathfrak{q}_2 \in \operatorname{spec} B$, which contracts to $\mathfrak{p}_2/\mathfrak{p}_1$ in $A/\mathfrak{p}_1$.

Then $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$, by the following diagram:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ A/\mathfrak{p}_1 & \longrightarrow & B/\mathfrak{q}_1 \end{array}$$

Since $\mathfrak{q}_2/\mathfrak{q}_1$ lies in $B/\mathfrak{q}_1$, and this contracts to $\mathfrak{p}_2/\mathfrak{p}_1$ in $A/\mathfrak{p}_1$, which contracts to $\mathfrak{p}_2$ in A. Then going the other way, we get $\mathfrak{q}_2/\mathfrak{q}_1$ contracts to $\mathfrak{q}_2$ in B. And since the diagram commutes, $\mathfrak{q}_2$ contracts to $\mathfrak{p}_2$ in A. □

Proposition 10.4 (Going down)**.** *Let $A \subseteq B$ be an integral extension of integral domains. Assume that A is integrally closed (i.e. integrally closed in* $\mathrm{Frac}(A)$*). Then if $\mathfrak{p}_1 \supseteq \mathfrak{p}_2$ in $\operatorname{spec} A$ and $\mathfrak{q}_1 \in \operatorname{spec} B$. Then there exists $\mathfrak{q}_2 \in \operatorname{spec} B$ with $\mathfrak{q}_1 \supseteq \mathfrak{q}_2$ such that $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$.*

Proof. Consider

$$A \to B \to B_{\mathfrak{q}_1}$$

The first map is injective, since it is inclusion, and the second is injective since a localisation of an integral domain is injective. Therefore the overall map is also injective. We want to prove: $\exists \mathfrak{n} \in \operatorname{spec} B_{\mathfrak{q}_1}$ such that $\mathfrak{n} \cap A = \mathfrak{p}_2$.

This is enough, since then we have $(\mathfrak{n} \cap B) \cap A = \mathfrak{p}_2$. But then $\mathfrak{n} \cap B$ is a contraction of a prime ideal, so it must be in $\operatorname{spec} B$. So what we really want is:

Want: $(\mathfrak{p}_2 B_{\mathfrak{q}_1}) \cap A \subseteq \mathfrak{p}_2$. (Since $\supseteq$ always holds[13].) It's useful to know that $(\mathfrak{p}_2 B_{\mathfrak{q}_1}) = (\mathfrak{p}_2 B) B_{\mathfrak{q}_1}$.

So take $\frac{y}{s} \in \mathfrak{p}_2 B_{\mathfrak{q}_1} \cap A$ where $y \in \mathfrak{p}_2 B$ and $s \in B \setminus \mathfrak{q}_1$. Since this is what a general element looks like in $\mathfrak{p}_2 B_{\mathfrak{q}_1}$.

Now we have that $A \subseteq B$ is an integral extension, so the integral closure of $\mathfrak{p}_2$ in B is $\sqrt{frakp_2 B}$. Therefore y is integral over $\mathfrak{p}_2$. We also have A is integrally closed over $\mathrm{Frac}(A)$, and that $\mathfrak{p}_2 = \sqrt{\mathfrak{p}_2}$ so the minimal equation of $y \in \mathrm{Frac}(B)$ over $\mathrm{Frac}(A)$ has the form

$$y^r + u_1 y^{r-1} + \cdots + u_r y^0 = 0 \tag{$*$}$$

where each $u_i \in \mathfrak{p}_2$. Then I can write

$$y = \frac{y}{s} \cdot s$$

where $y \in B$, $s \in B$, and $\frac{y}{s} \in A$. So, in particular $y, s \in \mathrm{Frac}(B)$ and $\frac{y}{s} \in \mathrm{Frac}(A)$. Then rewriting $(*)$:

$$\left(\frac{y}{s} \cdot s\right)^r + u_1 \cdot \left(\frac{y}{s} \cdot s\right)^{r-1} + \cdots + u_r \cdot \left(\frac{y}{s} \cdot s\right)^0$$

Then we multiply through by $\left(\frac{s}{y}\right)^r$ to get

$$s^r + \left(\frac{s}{y}\right)^1 \cdot u_1 \cdot s^{r-1} + \cdots + \left(\frac{s}{y}\right)^r \cdot u_r \cdot s^0 = 0 \tag{$**$}$$

[13] since if an ideal is contacted, and you extend it and contract it again, you get back to the same place.

Recall again that the $u_i \in \mathfrak{p}_2$, $s \in B$, and $\frac{s}{y} \in \mathrm{Frac}(A)$.

Now, since $s \in B$, s is integral over A. Therefore, the coefficients in $(**)$ are in A. So

$$\left(\frac{s}{y}\right) \cdot u_1, \ldots, \left(\frac{s}{y}\right)^r \cdot u_r \in A$$

Now assume that $\frac{y}{s} \notin \mathfrak{p}_2$. Then

$$u_i = \left(\frac{y}{s}\right)^i \cdot \left(\frac{s}{y}\right)^i \cdot u_i.$$

Then $\left(\frac{y}{s}\right)^i$ is in A, but not in $\mathfrak{p}_2$. We just concluded that $\left(\frac{s}{y}\right)^i \cdot u_i \in A \cap \mathfrak{p}_2$ since we know that $u_i \in \mathfrak{p}_2$. Therefore, by $(**)$, $s^r \in \mathfrak{p}_2 B$, since $\left(\frac{s}{y}\right)^i \cdot u_i \in \mathfrak{p}_2$. But

$$\mathfrak{p}_2 B \subseteq \mathfrak{p}_1 B = (\mathfrak{q}_1 \cap A)B \subseteq \mathfrak{q}_1.$$

So $s \in \mathfrak{q}_1$. But we *know* that $s \notin \mathfrak{q}_1$, since we divide by it in the localization ring $B_{\mathfrak{q}_1}$, so it precisely cannot be in $\mathfrak{q}_1$. Contradiction. □

Remark. If there is a finitely algebra over a field k which is also an integral domain, then we know it contains a polynomial algebra (and is integral over it). A polynomial algebra is always integrally closed. So this is precisely a time when we can use the going down theorem.

11 Primary decomposition of ideals

Definition 11.1 (Primary ideal)**.** Let I be an ideal of R.

1. I is prime if $R/I \neq 0$ and 0 is the only 0 divisor.
2. I is radical if in R/I, the only nilpotent element is 0.
3. I is **primary** if $R/I \neq 0$ and every zero divisor in R/I is nilpotent.

So, I is prime $\Rightarrow$ I is radical and primary.

(R itself is radical, but not prime or primary.)

Example. Let $R = \mathbb{Z}$, then (6) is radical, but not primary, because in $\mathbb{Z}/(6)$, $2 \cdot 3 = 0$ but neither 2 nor 3 are nilpotent.

(9) is primary but not radical, $R/(9)$. So in general, if $0 \neq x \in \mathbb{Z}$, then (x) is prime iff $|x|$ is prime, and (x) is radical iff x is square free. (x) is primary iff $x = p^n$ for some p prime, $n \geqslant 1$.

Proposition 11.2. *Let $I \subsetneq R$ an ideal, then*

1. *I primary $\Rightarrow \sqrt{I} = \mathfrak{p}$ prime. We say I is $\mathfrak{p}$ primary. (So)*
2. *$\mathfrak{q}_1, \dots, \mathfrak{q}_n$ are $\mathfrak{p}$-primary, then $\mathfrak{q}_1 \cap \dots \cap \mathfrak{q}_n$ is also $\mathfrak{p}$-primary.*
3. *If I has a **primary decomposition***

$$I = \mathfrak{q}_1 \cap \dots \cap \mathfrak{q}_n, \quad \mathfrak{q}_i \text{ primary.} \tag{$*$}$$

   *Then I has a **minimal primary decomposition** of the same form as $(*)$, such that $\sqrt{q_1}, \dots, \sqrt{q_i}$ are distinct, and no $\mathfrak{q}_i$ can be dropped*

4. *If $\sqrt{I} \in \operatorname{mspec} R$, then I is primary.*
5. *If R is noetherian, then every ideal I has a primary decomposition.*

Proof. In example sheet

Example. In $\mathbb{Z}$, $(90) = (2) \cap (3^2) \cap (5)$ is a primary decomposition.

Examples. For $p \in \operatorname{spec} R$, if $\mathfrak{p}^n$ is primary, then $\mathfrak{p}^n$ is $\mathfrak{p}$ primary, since $\sqrt{\mathfrak{p}^n} = \mathfrak{p}$.

1. Not every primary ideal is a power of a prime ideal. For example, let

$$R = k[X, Y] \qquad\qquad \mathfrak{q} = (X, Y^2)$$

   Then $\mathfrak{q}$ is primary, since $\sqrt{\mathfrak{q}} = (X, Y) \in \operatorname{mspec} R$. So $\mathfrak{q}$ is (X, Y)-primary.

   Another way to see that $\mathfrak{q}$ is primary is by looking at

$$k[X, Y]/(X, Y^2) \cong k[Y]/(Y^2)$$

   If $f \in k[Y]$ and $f + (y^2)$ is a zero-divisor, then $y \sim f$, so $f + (y^2)$ is nilpotent. If $\mathfrak{q} = p^n$ for some $\mathfrak{p} \in \operatorname{spec} k$, then

$$(X, Y) = \sqrt{\mathfrak{q}} = \sqrt{\mathfrak{p}^n} = \mathfrak{p}.$$

   Then $(X, Y) = \sqrt{\mathfrak{q}} = \sqrt{\mathfrak{p}^n} = \mathfrak{p}$, so we have

$$(X, Y) \supsetneq (X, Y^2) \supsetneq (X, Y)^2.$$

   So $(X, Y^2) \neq (X, Y)^n$ for any $n \geqslant 1$.

2. A power of a prime ideal, $\mathfrak{p}^n$ is not always primary. Let

$$R = k[X, Y, Z]/(XY - Z^2) = k[\bar{X}, \bar{Y}, \bar{Z}],$$

   where $\bar{X}$ is the image of X under this quotient. Let $\mathfrak{p} = (\bar{X}, \bar{Z})$. Then we claim that $\mathfrak{p}$ is prime, since $R/\mathfrak{p} \cong k[Y]$ (since X and Z are both sent to

zero in the quotient). Therefore $\mathfrak{p}$ is prime. However $\mathfrak{p}^2 = (\bar{X}^2, \bar{X}\bar{Z}, \bar{Z}^2)$, so $\bar{X} \cdot \bar{Y} = \bar{Z}^2 \in \mathfrak{p}^2$. So $(\bar{X} + \mathfrak{p}^2) \cdot (\bar{Y} + \mathfrak{p}^2) = 0 + \mathfrak{p}^2$. But $X + \mathfrak{p}^2 \neq 0 + \mathfrak{p}^2$, so $\bar{Y}$ is indeed a zero-divisor in $R/\mathfrak{p}^2$. But it is not nilpotent, since

$$R/\mathfrak{p}^2 = \frac{k[X, Y, Z]}{XY, X^2, XZ, Z^2}$$

and so no power of Y ends up in $\mathfrak{p}$.

Theorem 11.3. *Let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$, a minimal primary decomposition. Now let $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$. Then:*

1. *The associated primes $\mathfrak{p}_1, \ldots, \mathfrak{p}^n$ are determined by I. (So there can be multiple minimal primary decompositions, but the set of radicals does depend only on I).*
2. *The **isolated primes** of I: the minimal elements among $\mathfrak{p}_1, \ldots \mathfrak{p}_n$ when orderd by inclusion are exactly the minimal prime ideals of R containing I. These are called the **isolated primes** of I.*
3. *The **embedded primes** of I are the associated primes which are not isolated (i.e. the rest of them).*
4. *If $\mathfrak{p}_1, \ldots, \mathfrak{p}_t$, with $t \leqslant n$ are the isolated primes of I, then $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ are determined only by I.*

Proof. In example sheet.

Example. Let $R = k[X, Y]$, and $I = (X^2, XY)$, then

$$I = (X) \cap (X, Y)^2,$$

and

$$I = (X) \cap (X^2, Y).$$

$$\sqrt{(X)} = X \qquad \sqrt{(X, Y)^2} = (X, Y) \qquad \sqrt{(X^2, Y)} = (X, Y)$$

The associated primes of I are therefore (X) and (X, Y). We have $(X) \subseteq (X, Y)$, and (X) is an isolated prime, and (X, Y) is an embedded prime.

If $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ is a minimal primary decomposition, then $\sqrt{\mathfrak{q}_1}, \ldots, \sqrt{\mathfrak{q}_t}$ are the isolated primes. Then $\sqrt{I} = \sqrt{q_1} \cap \cdots \cap \sqrt{\mathfrak{q}_n}$[14] is a minimal primary decomposition of the radical of $\sqrt{I}$, and all associated primes are isolated. So the minimal primary decomposition of a radical ideal completely determines the minimal primary decomposition. So going from I to $\sqrt{I}$ means that we

[14] Because these contain all the other ideals.

"only remember the isolated primes of I," and forget all the other embedded ideals.

Example. In $k[T_1, \ldots, T_n]$ where $k \subseteq \mathbb{C}$. Then

$$V(I) = V(\sqrt{I})$$

and

$$I(V(I)) = \sqrt{I},$$

so $V(I)$ remembers $\sqrt{I}$ and nothing else about I.

12 Direct and inverse limits and completions

Let $\mathcal{C}$ be a category (like sets, groups, rings, R-modules, or R algebras).

Definition 12.1 (Directed Set)**.**

1. A **directed set** is a partially ordered set such that

$$\forall a, b \in I \ \exists c \in I \text{ s.t. } a \leqslant c, \ b \leqslant c$$

2. A **direct system** on I is

$$((X_i)_{i \in I}, (f_{ij})_{i \leqslant j})$$

where the X_i are objects, and the f_{ij} are morphisms, satisfying:

- $f_{ii} = \mathrm{Id}_{X_i}$ for all $i \in I$.
- $f_{jk} \circ f_{ij} = f_{ik}$ for all $i \leqslant j \leqslant k$ in I.

3. An **inverse system** on I is

$$((Y_i)_{i \in I}, (h_{ij})_{i \leqslant j})$$

where $h_{ij} : X_j \to X_i$, which satisfies:

- $h_{ii} = \mathrm{Id}_{X_i}$
- $f_{ij} \circ f_{jk} = f_{ik}$ for all $i \leqslant j \leqslant k$ in I.

Example. Let $I = (\mathbb{N}, \leqslant)$

1. Define a direct system on I which is given by first fixing a prime p, then take

$$X_i = \mathbb{F}_{p^{i!}} \qquad\qquad f_{ij} : \mathbb{F}_{p^{i!}} \to \mathbb{F}_{p^{j!}}$$

Recall that if $a \sim b$, then there is an embedding $\varphi : \mathbb{F}_{p^a} \hookrightarrow \mathbb{F}_{p^b}$, and the collection of embeddings $\mathbb{F}_{p^a} \hookrightarrow \mathbb{F}_{p^b}$ is given by

$$\{x \mapsto (\varphi(x))^{p^c} \text{ s.t. } 0 \leqslant c \leqslant a-1\}$$

So for $i \leqslant j$, we will define

$$f_{ij} f_{j-1,j} \circ \cdots \circ f_{i,i+1}.$$

2. Define an inverse system on I by

$$Y_i = \mathbb{Z}/p^i\mathbb{Z} \qquad\qquad f_{ij} : \mathbb{Z}/p^j\mathbb{Z} \to \mathbb{Z}/p^i\mathbb{Z}$$

where f_{ij} is given by the natural projection.

Definition 12.2 (Direct limit/Inverse limit)**.** Let $(I, \leqslant)$ be a directed set, then

1. If $((X_i), (f_{ij})_{i \leqslant j}) := D$ is a direct set on I. The **direct limit** of D is denoted
$$\varinjlim X_i = \left(\bigsqcup_{i \in I} X_i \right) / \sim$$
where for $x_i \in X_i$, and $x_j \in X_j$, we have
$$x_i \sim x_j \Leftrightarrow f_{ik}(x_i) = f_{jk}(x_j)$$
for some $i \leqslant k$, $j \leqslant k$. (Equivalently, we could have written "the smallest equivalence relation satisfying $x_i \sim f_{ij}(x_i)$" This gives more of a sense of what we are trying to capture.)
2. Let $((Y_i)_{i \in I}, (h_{ij})_{i \leqslant j}) := E$ be an inverse system on I, then the **inverse limit** of E is:
$$\varprojlim = \{\bar{y} \in \prod_{i \in I} Y_i \mid y_i = h_{ij}(y_j) \ \ \forall i \leqslant j\}$$

Examples. 1. The direct limit $\varinjlim \mathbb{F}_{p^{i!}}$ is an algebraic closure of $\mathbb{F}_p$.

Proof. First, $\mathbb{F}_p^{\text{alg}} := \varinjlim \mathbb{F}_{p^{i!}}$ is algebraic over $\mathbb{F}_p$. Take $[x] \in \mathbb{F}_p^{\text{alg}}$, so then $x \in \mathbb{F}_{p^{i!}}$ for some $i \geqslant 1$. Then

$$x^{p^{i!}} - x = 0$$

Therefore

$$[x]^{p^{i!}} - [x] = 0,$$

so we are done, since there is a polynomial equation with coefficients in $\mathbb{F}_p$ which has $[x]$ as a solution.

Now we show that $\mathbb{F}_p^{\text{alg}}$ is algebraically closed. So take

$$[h] \in (\mathbb{F}_p^{\text{alg}})[T], \quad h \in \mathbb{F}_{p^{i!}}[T]$$

where $[h]$ is an abuse of notation, meaning we take the equivalence class of all the coefficients. Therefore h splits under some extension of $\mathbb{F}_{p^{i!}} \to \mathbb{F}_{p^l}$, but then $\mathbb{F}_{p^l}$ embeds into $\mathbb{F}_{p^{l!}}$, so h splits in $\mathbb{F}_{p^{l!}}$. So h must split under $h_{ij} : \mathbb{F}_{p^{i!}} \to \mathbb{F}_{p^{j!}}$. Then also $[h]$ must split in $\mathbb{F}_p^{\text{alg}}$.

Let $\mathbb{Z}_p := \varprojlim \mathbb{Z}/p^i\mathbb{Z}$, where $\mathbb{Z}_p$ is the ring of p-adic integers. For example, take $p = 5$, and a sequence $(d_i)_{i=0}^{\infty}$ where $0 \leqslant d_i \leqslant 4$. Then $1 \in \mathbb{Z}_5$ is:

$$(1 + 5\mathbb{Z}, 1 + 5^2\mathbb{Z}, 1 + 5^3\mathbb{Z}, \dots),$$

and $-1 \in \mathbb{Z}_5$ is

$$(4_5 + 5\mathbb{Z}, 44_5 + 5^2\mathbb{Z}, 444_5 + 5^3\mathbb{Z}, \dots)$$

where we write numbers in base 5.

Definition 12.3 ($\mathfrak{a}$-adic completion)**.** Let R be a ring, and $\mathfrak{a} \subseteq R$ an ideal, then the **$\mathfrak{a}$-adic completion** of R is

$$\hat{R} = \varprojlim R/\mathfrak{a}^i,$$

where our direct system if $R/\mathfrak{a}, R/\mathfrak{a}^2, R/\mathfrak{a}^3, \dots$, where the maps map from higher powers of $\mathfrak{a}$ to lower powers of $\mathfrak{a}$.

Examples.

1. If $R = \mathbb{Z}$ and $\mathfrak{a} = (p)$, then
$$\hat{R} = \mathbb{Z}_p$$
2. If $R = k[T]$, and $\mathfrak{a} = (T)$, then
$$\hat{R} = \varprojlim R/(T^i),$$
so in this ring we don't have to stop polynomials at any point, so this is given by
$$\hat{R} = k[[T]] = \{\sum_{n=0}^{\infty} a_i T^i \mid a_k \in k\}.$$

Definition 12.4 ($\mathfrak{a}$-adic completion of a module)**.** If M is an R-module, and $\mathfrak{a} \subseteq R$ an ideal, then the **$\mathfrak{a}$-adic completion of** M is

$$\hat{M} = \varprojlim M/\mathfrak{a}^i M$$

which is an $\hat{R}$-module.

Definition 12.5 (Filtration)**.**

More generally, a **filtration** of an R-module M is a sequence $(M_n)_{n\geqslant 1}$ of submoudles of M such that $M_n \supseteq M_{n+1}$.

The completion of M with respect to $(M_n)_{n\geqslant 1}$ is

$$\lim_{\leftarrow} M/M_n$$

Theorem 12.6. *Let R be a noetherian ring, and $\mathfrak{a} \subseteq R$ an ideal, then:*

1. *$\hat{R}$ is noetherian.*
2. *$\hat{R} \otimes_R (\cdot)$ is exact.*
3. *If M is a finitely generated R-module, then*

$$\hat{R} \otimes_R M \to \hat{M}$$

is an $\hat{R}$-linear isomorphism.